7. Environmental Impact of Aviation



7. Environmental Impact of Aviation

7.1 Aircraft Noise

7.2 Aircraft Exhaust Emissions

Overview: Environmental impact of aviation

Aviation

Noise

Emissions

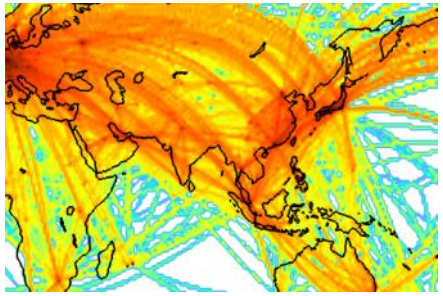
local

local

global







Health of population

Local air quality

Climate change

Environmental impact

Regulations, Charges, external costs

7. Environmental Impact of Aviation

7.1 Aircraft Noise



Definitions (1)

- Sound waves are **oscillations of the air** that can be heard by the human ear or measured by a physical device.

- Sound progresses as a **periodic oscillation** of the air density through **longitudinal waves**.

- Sound waves are characterized by a **local speed of sound and a local sound pressure**.

- The sound pressure amplitude ‘p’ of the sound wave can be used to measure the **loudness**:
  - Lowest audible pressure: $2 \cdot 10^{-5}$ Pa
  - Pain starts at: $2 \cdot 10^{2}$ Pa

Definitions (2)

- The sound pressure is usually given through the **Sound Pressure Level (SPL)**

$$SPL = 20 \log \frac{p}{p_0} [dB] \text{ with } p_0 = reference\ pressure = 2 \cdot 10^{-5} Pa$$

- The **Sound Intensity (I)** is a quantitative physical metrics to address the energy transport of a sound wave:

$$I = \frac{p^2}{\rho \cdot c} = \frac{Power}{Area} \left[\frac{W}{m^2}\right]$$

$$\text{Level of Intensity: } L_I = 10 \log \frac{I}{I_0} [dB]$$

with ρ = Air density

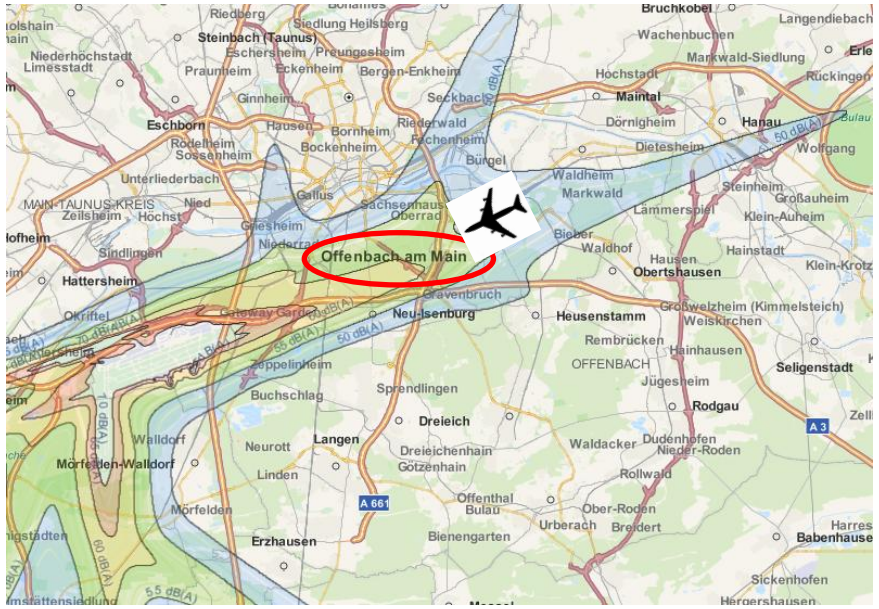
c = Speed of sound

I_0 = Reference Intensity = 10^{-12} W

Exercise: Aircraft Noise



Height reference (Offenbach town)	100ft
SPL_ref	100 dB(A)
Flight altitude (glide slope angle: 3,0°)	2770ft
Flight altitude (glide slope angle: 3,2°)	2930ft
Offenbach am Main (height)	400ft



A Boeing 747-8i flies over Offenbach am Main (reference point) during its final approach to Frankfurt.
What aircraft noise residents must expect for two different approach procedures?

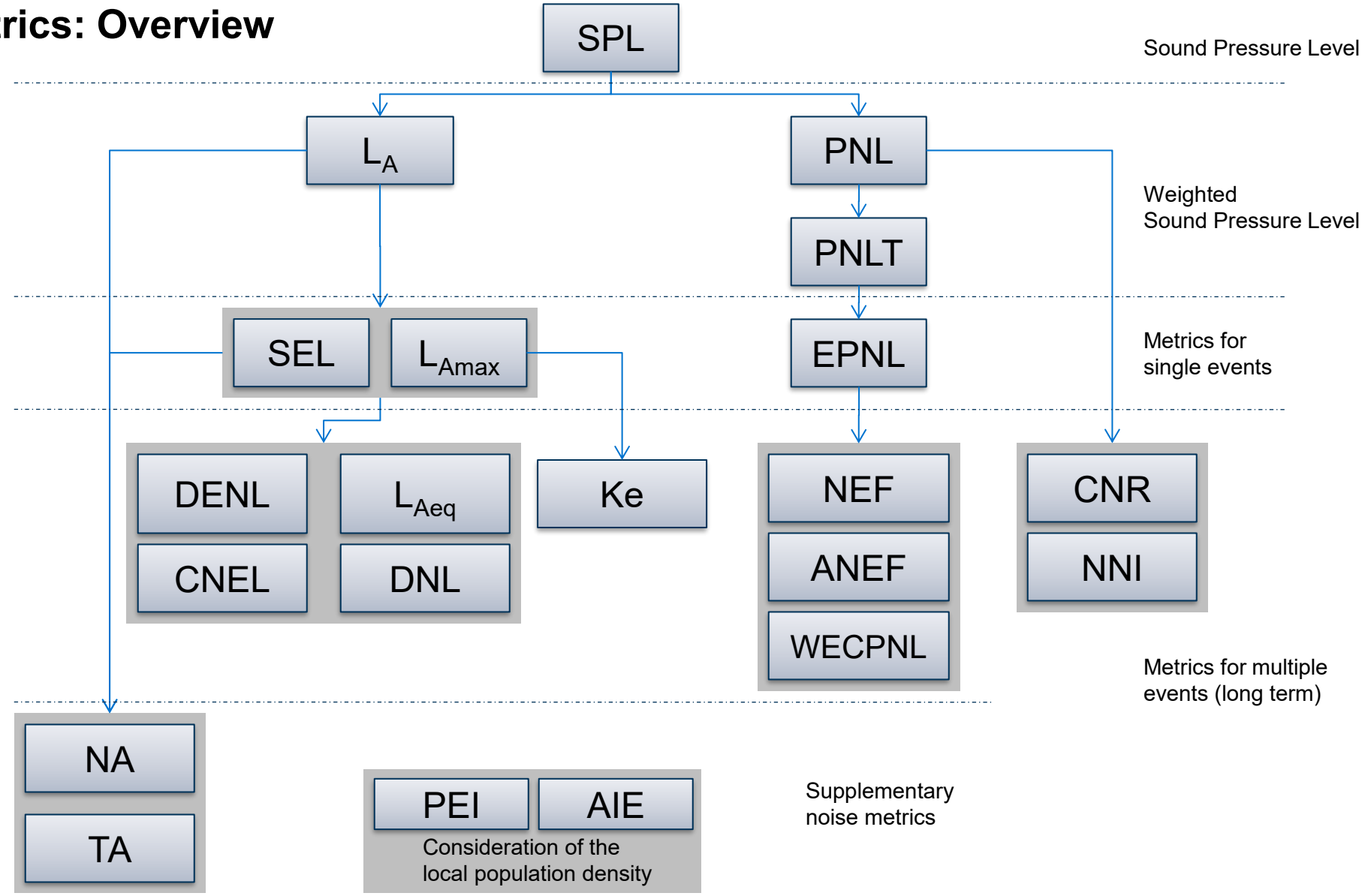
$$SPL_{3.0°} = SPL_{ref} - 20 \cdot log\left(\frac{h}{h_{ref}}\right) = 72.5\ dB(A)$$

$$SPL_{3.2°} = 71.9\ dB(A)$$

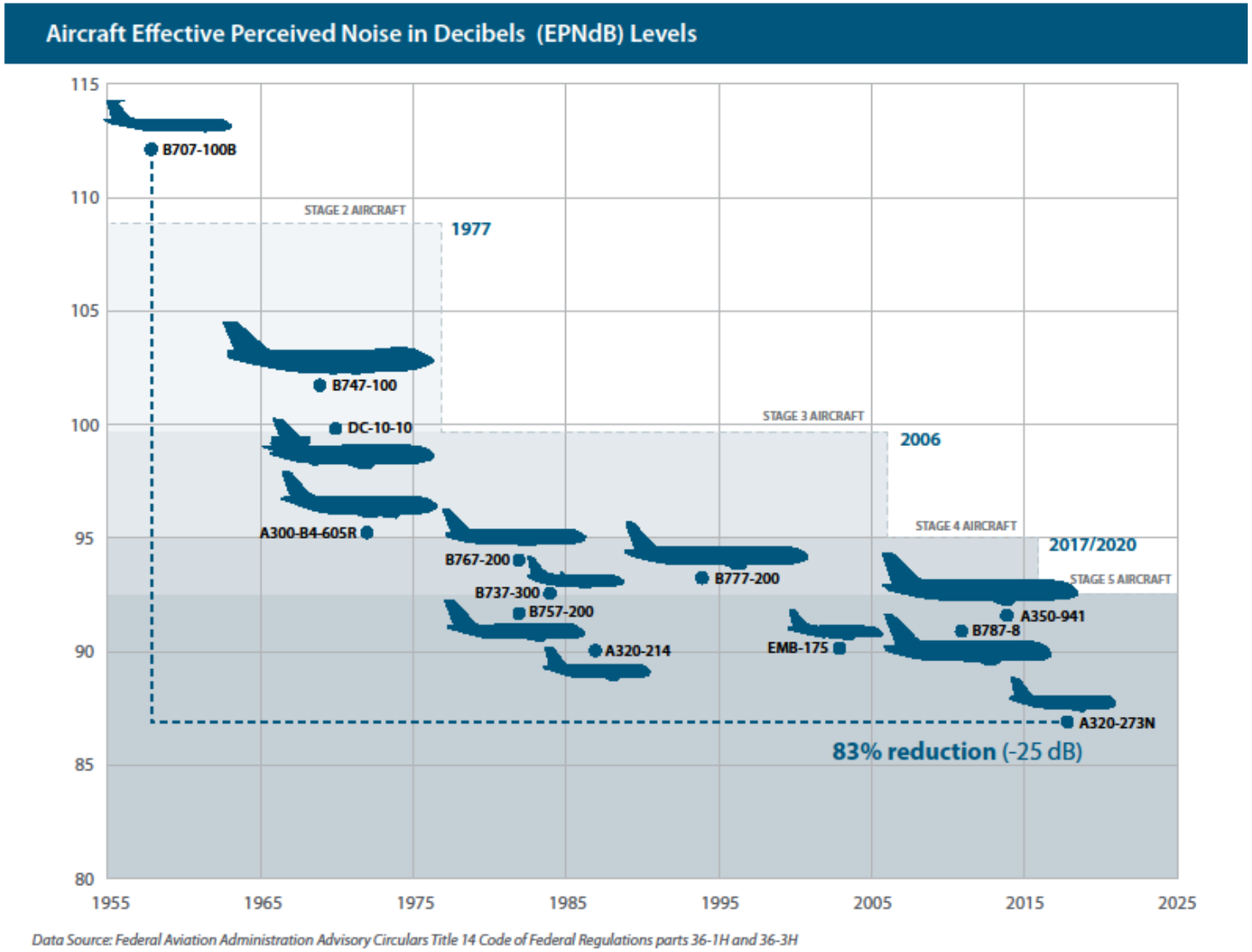
- Higher approach path would result in lower aircraft noise at ground

Picture source: framap.fraport.de; pinterest.ch; airwaysmag.com

Noise metrics: Overview



Noise metrics: Overview



Data Source: Federal Aviation Administration Advisory Circulars Title 14 Code of Federal Regulations parts 36-1H and 36-3H

Definitions (3)

Noise: subjective evaluation of the degree of annoyance of sound

- **Current maximum levels:**
  - **Perceived Noise Level (PNL):**

Measure of the current noise level, determined by empirical weighting of the intensities of multiple frequency bands relative to the overall noise level. This kind of frequency-based evaluation is exclusively used for aircraft noise.

  - **Tone Corrected Perceived Noise Level (PNLT):**

Addition of tone correction C for tone noise in the frequency spectrum:

PNLT = PNL + C

  - **A – Weighted Sound Pressure Level (L_A):**

Evaluation of the current noise level according to the A-weighting curve. This type of frequency-based evaluation is used for many kinds of noise sources.

Definitions (4)

- **Maximum level for one event:**
  - **Effective Perceived Noise Level (EPNL):**

  Measure of the noise level taking into account the chronological sequence of a sound event. Addition of a constant correction D through the integration of the current maximum levels PNLT during time. All levels are taken into account that exceed $PNLT_{MAX}$-10 dB:

$$EPNL = PNLT + D \text{ [EPNdB]}$$

- **Sound Exposure Level (SEL):**

Measure of the current noise level taking into account the chronological sequence of a sound event. Calculation analogous to EPNL, but without tone correction and taking the A-weighted Sound Pressure Level as a basis.

Definitions (5)

- **Level for multiple events:**
  - **Weighted Equivalent Continuous Perceived Noise Level (WECPNL):**

Measure of the annoyance of noise at an airport through the integration of measured EPNL during one year. This metrics is used rarely.

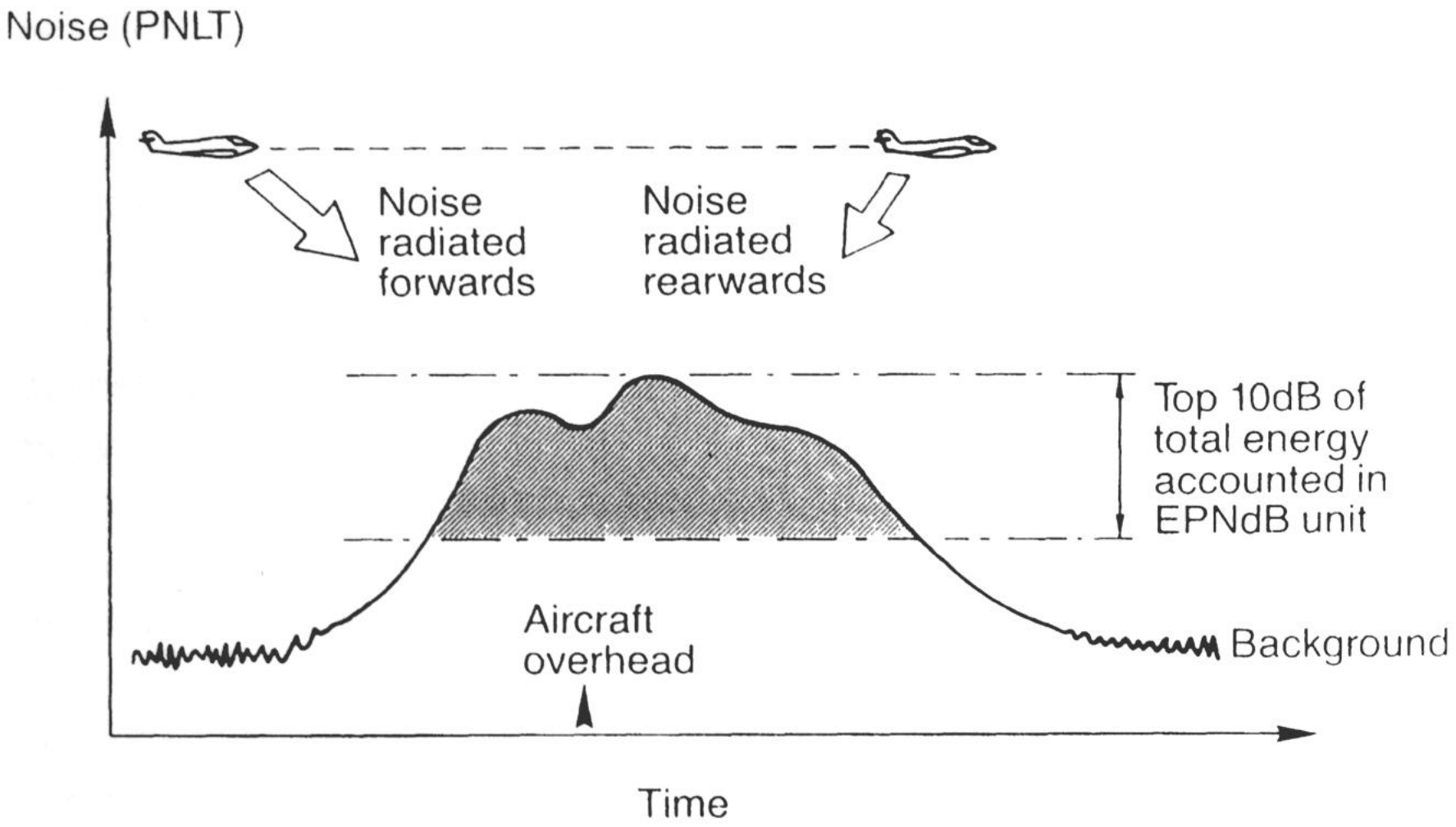
  - **Energy-equivalent Constant Sound Pressure (L_{eq}):**

Measure of the noise emission during one day on the basis of the SEL. This metrics is frequently used for noise assessments in Europe.

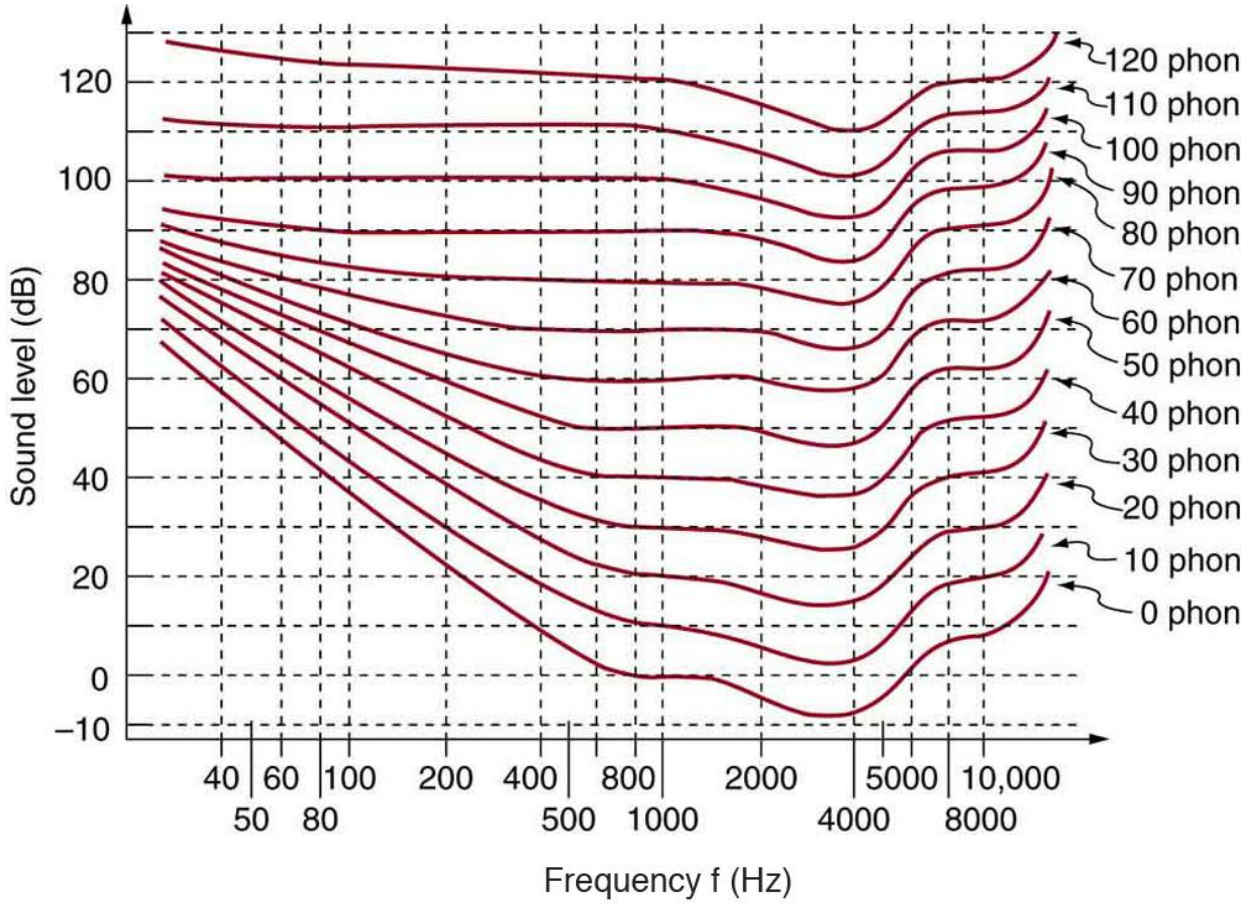
  - **Day-Night Level (DNL):**

Measure of the noise emission during one day on the basis of the SEL. This metrics is frequently used for noise assessments in North America. At night, noise events are evaluated with an addition of 10 dB relative to events during the day.

Determination of the EPNL



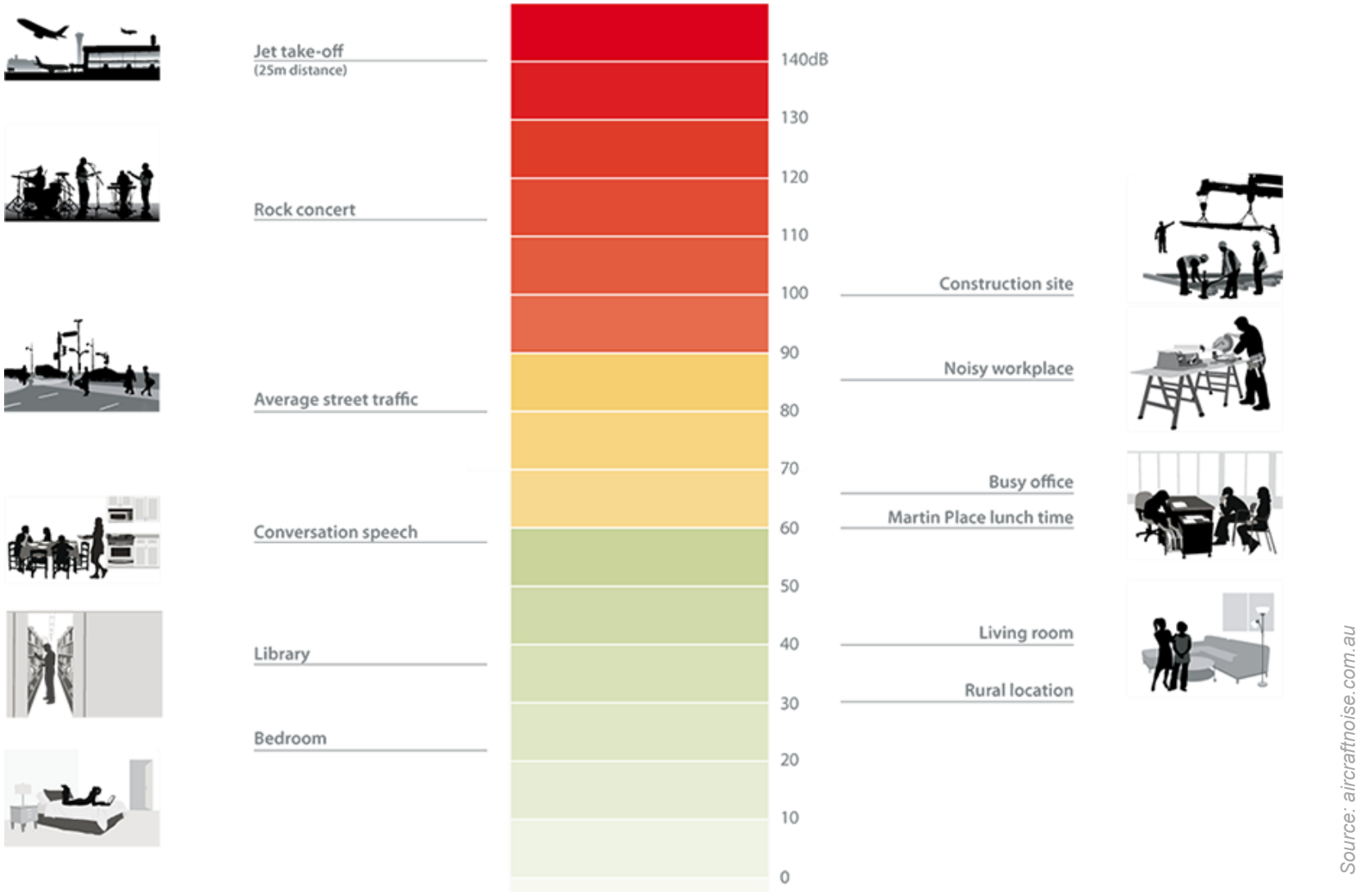
Relation between sound, frequency, and loudness



The diagram shows the relationship of loudness in phons to intensity level (in decibels) and intensity (in watts per meter squared) for persons with normal hearing. The curved lines are equal-loudness curves. All sounds on a given curve are perceived as equally loud. Phons and decibels are defined to be the same at 1000 Hz.

Image Source: Vietnam Open Educational Resources

Example noise levels



Regulations and restrictions

Two fundamental categories of regulations and restrictions can be distinguished:

- **Noise certification** of aircraft according to ICAO Annex 16
- **Operational restrictions** at airports

Noise-related laws and impacts

Time	Day	Night
Criterion	Constant sound level (L_{eq}, DNL, …)	Constant sound level and single events
Impacts	• Hearing impairment at L_{eq} > 75dB(A) • Increasing risk of heart attack • Communication disturbances at L_{eq} > 40dB(A), L_{Amax} = 55dB(A)	• Disturbances during sleep or while falling asleep • Waking up at levels L_{Amax} = 42dB(A) • Constant sound level: disturbed sleep at levels L_{eq} > 32dB(A) within sleeping room
Laws (Germany)	For existing airports: • L_{eq} = 60-65dB(A): Protection zone 2 • L_{eq} > 65dB(A): Protection zone 1 (for new or expanding airports, values are reduced by 5dB)	Authorized maximum (existing airports): • Single events: 6 x 57db(A) interior • Constant sound level: L_{eq} < 55 dB(A) exterior (for new or expanding airports 6 x 53 dB or 50dB, respectively)

Source: German Law for the Protection against Aircraft Noise; Bundesvereinigung gegen Fluglärm

Positioning of the microphones for noise certifications

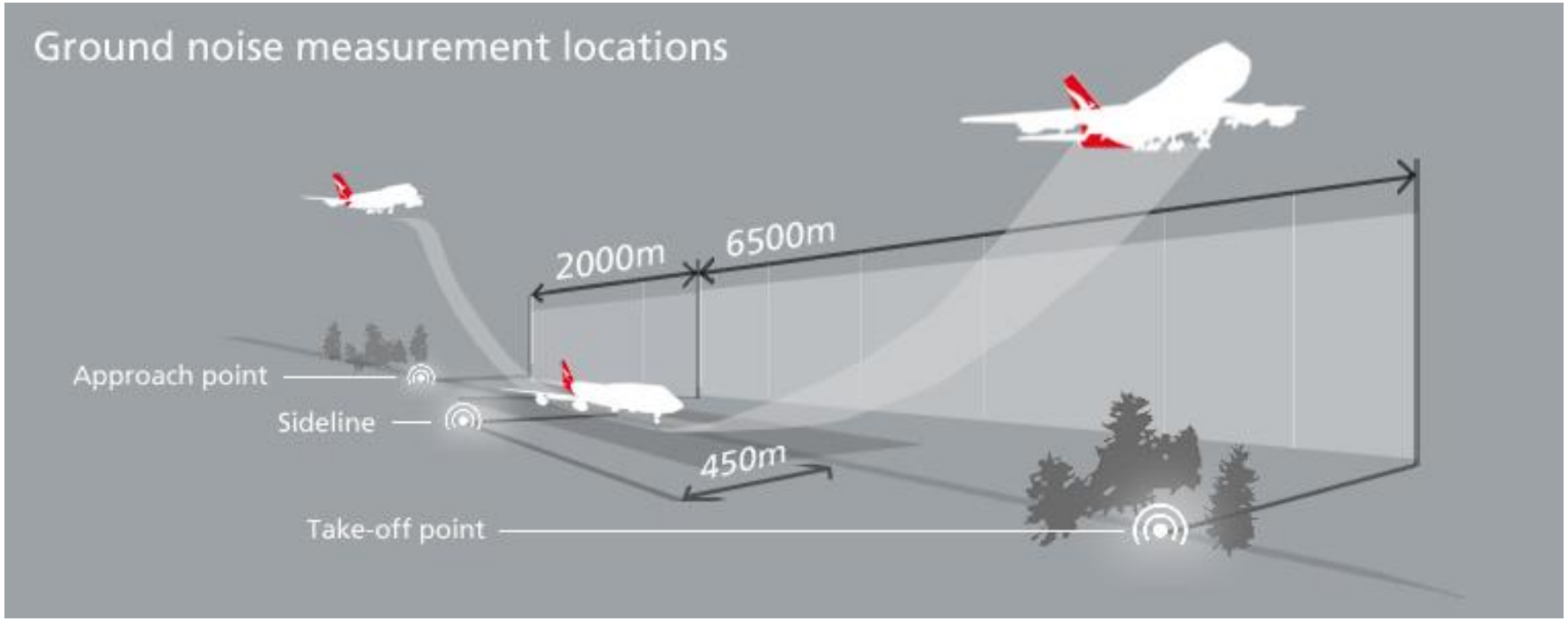


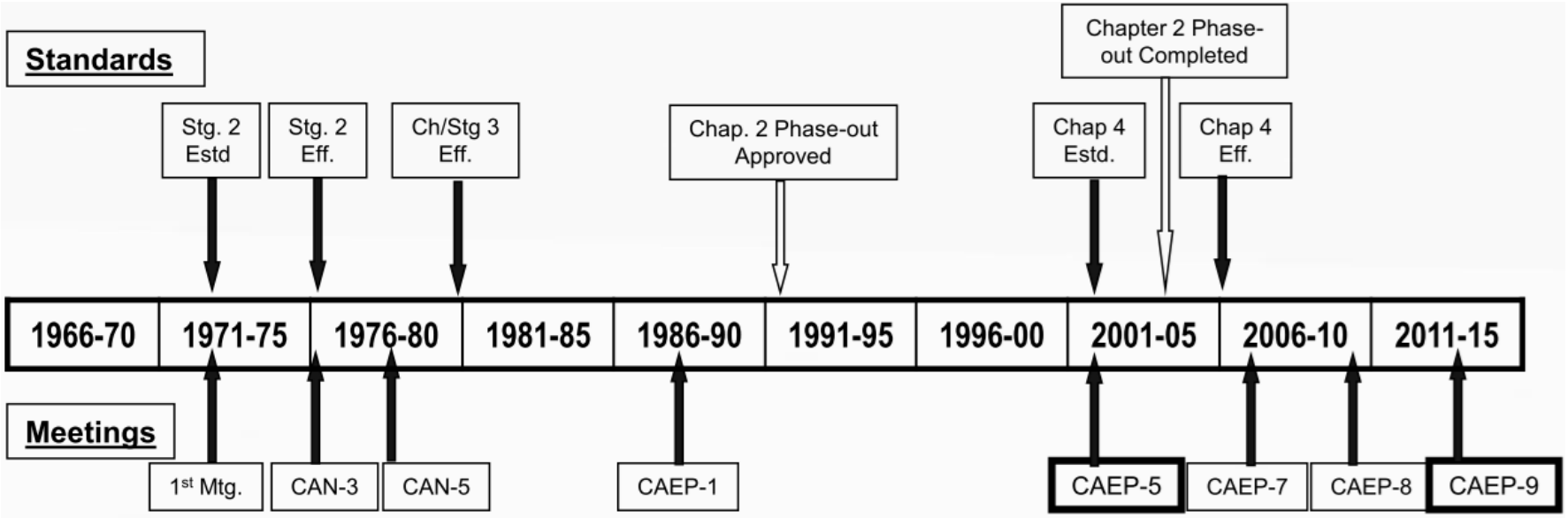
Image Source: Qantas

For aircraft noise certification, the sound pressure is measured at three positions:

- **Approach**: 2km from the runway threshold under the approach flight path
- **Sideline**: the highest noise measurement recorded at any point 450m from the runway axis during takeoff
- **Fly-over**: 6.5km from the brake-release point under the takeoff flight path

At each point of measurement, maximum allowable sound levels are defined.

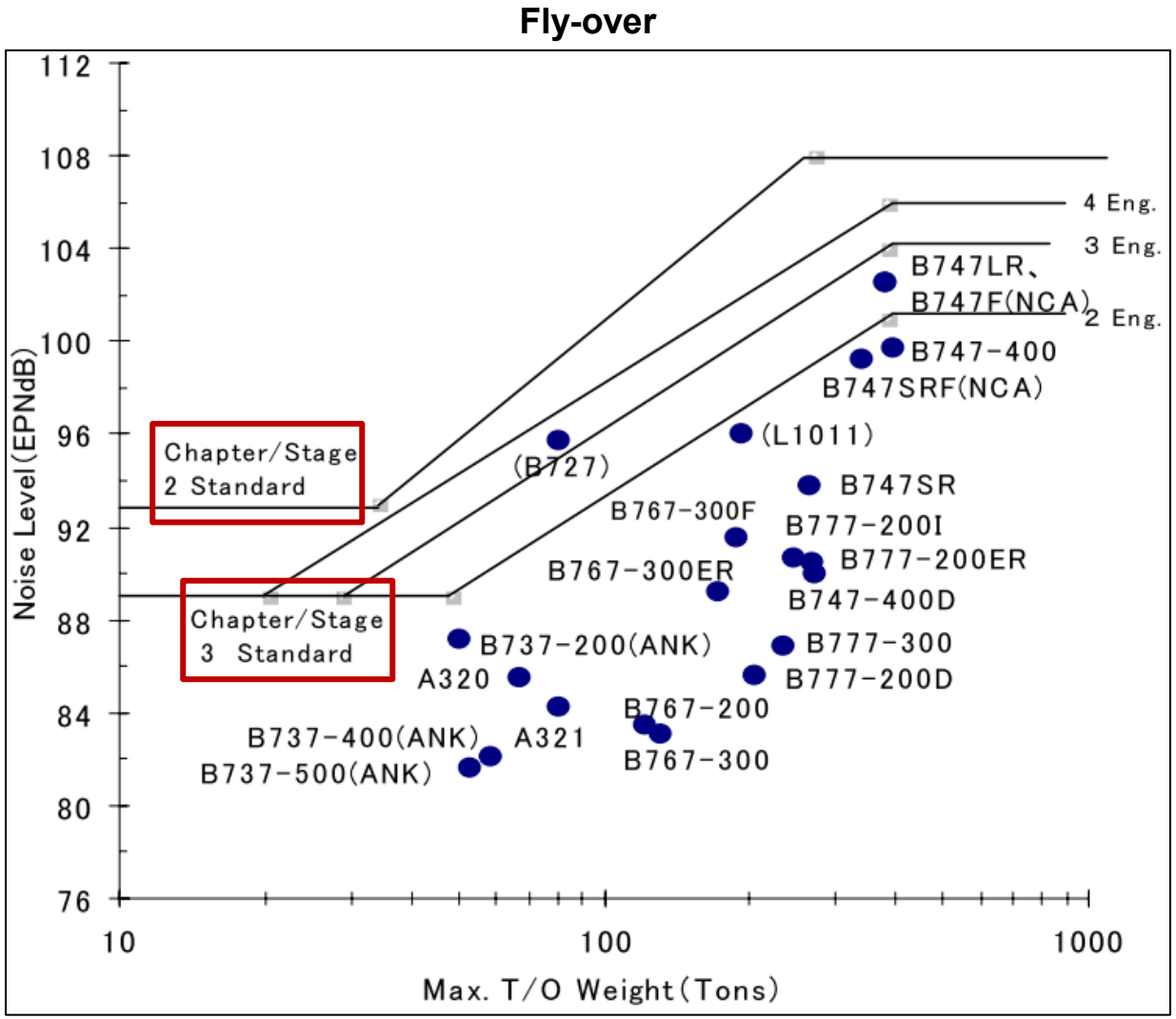
History of ICAO Noise Standards



Chapter	Applicable Year	Cum Margin (EPNdB)
2	1972	Ch. 3 ~-16
3	1978	0 (Ref)
4	2006	Ch. 3 +10

Source: ICAO

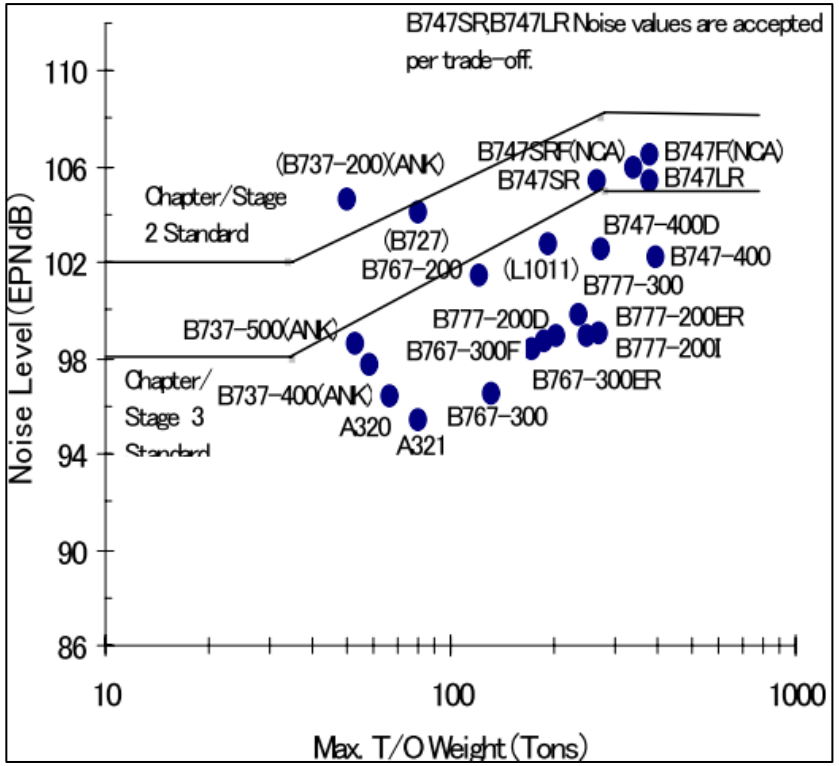
ICAO Noise Chapters 2 & 3



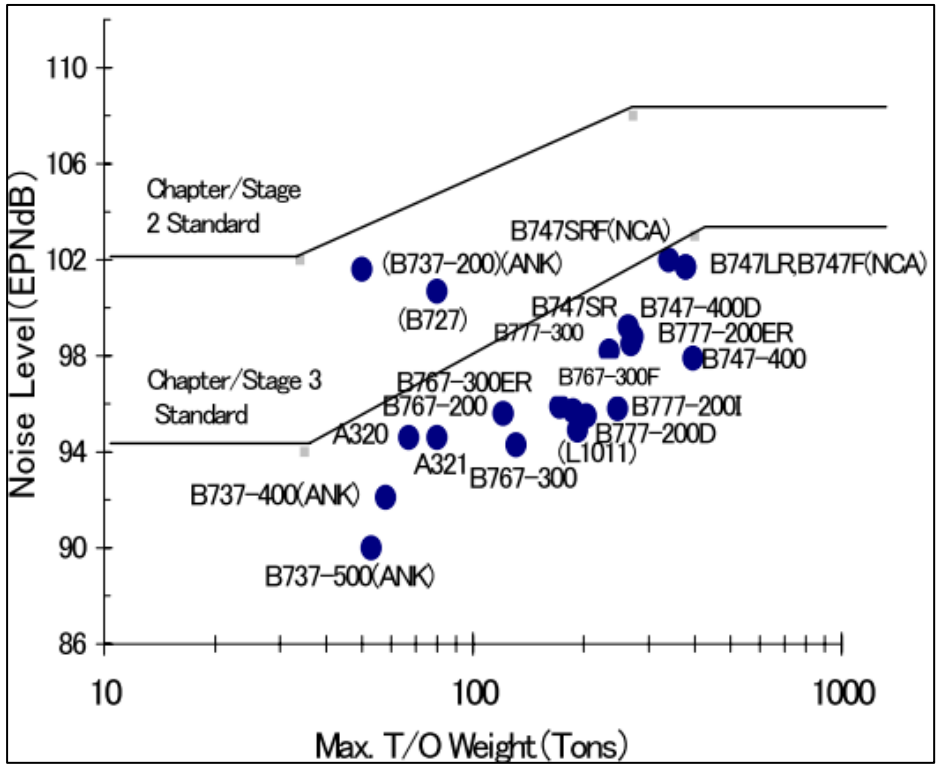
Source: ANA, ICAO Annex 16

ICAO Noise Chapters 2 & 3

Approach

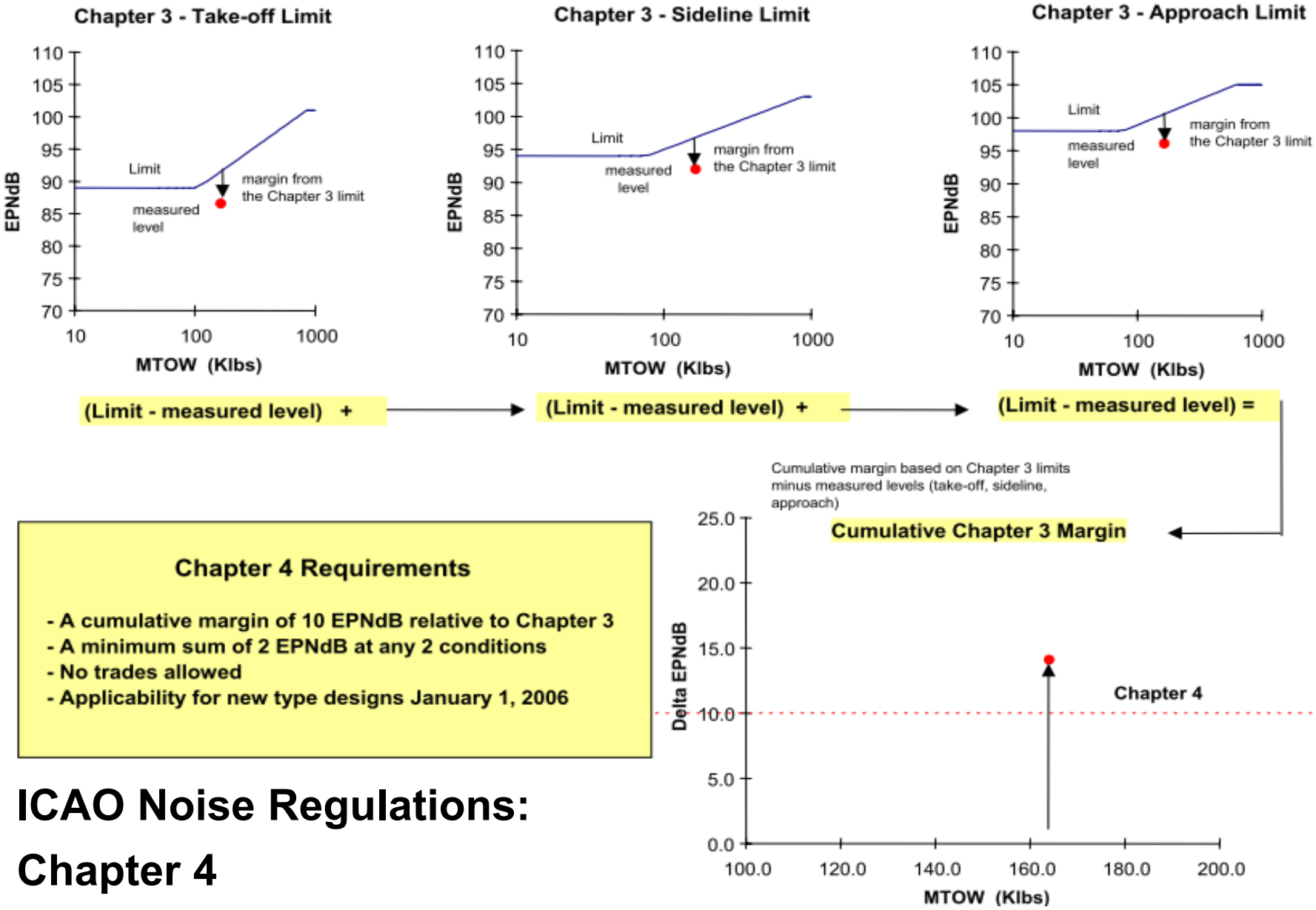


Sideline



Source: ANA, ICAO Annex 16

ICAO Noise Regulations: Chapter 4



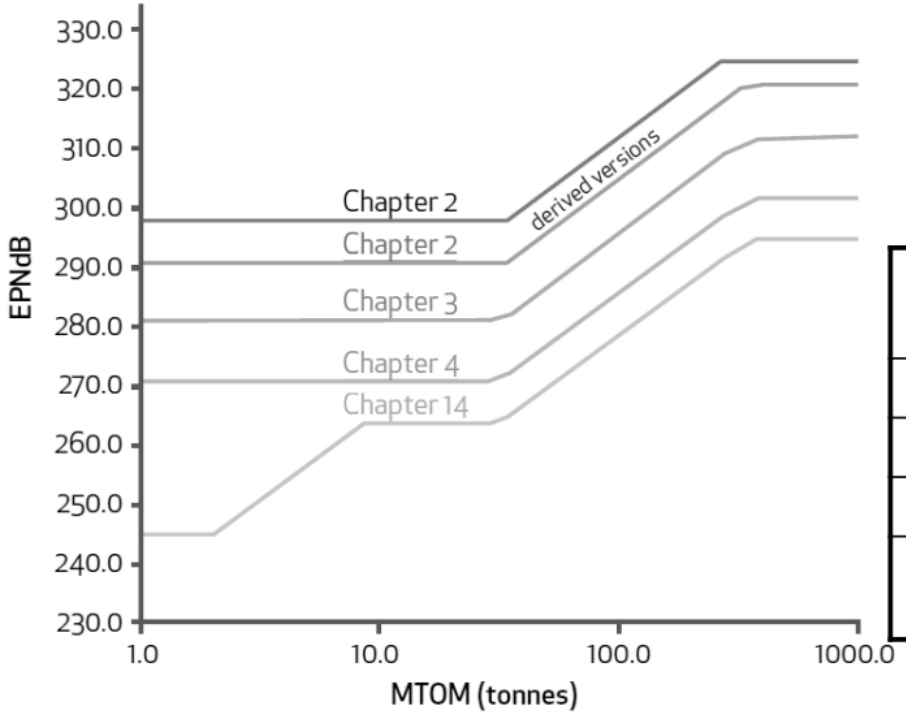
Chapter 4 Requirements

- A cumulative margin of 10 EPNdB relative to Chapter 3
- A minimum sum of 2 EPNdB at any 2 conditions
- No trades allowed
- Applicability for new type designs January 1, 2006

ICAO Noise Regulations: Chapter 14

CAEP agreed to recommend an amendment to Annex 16, Vol. I (new Chapter 14):

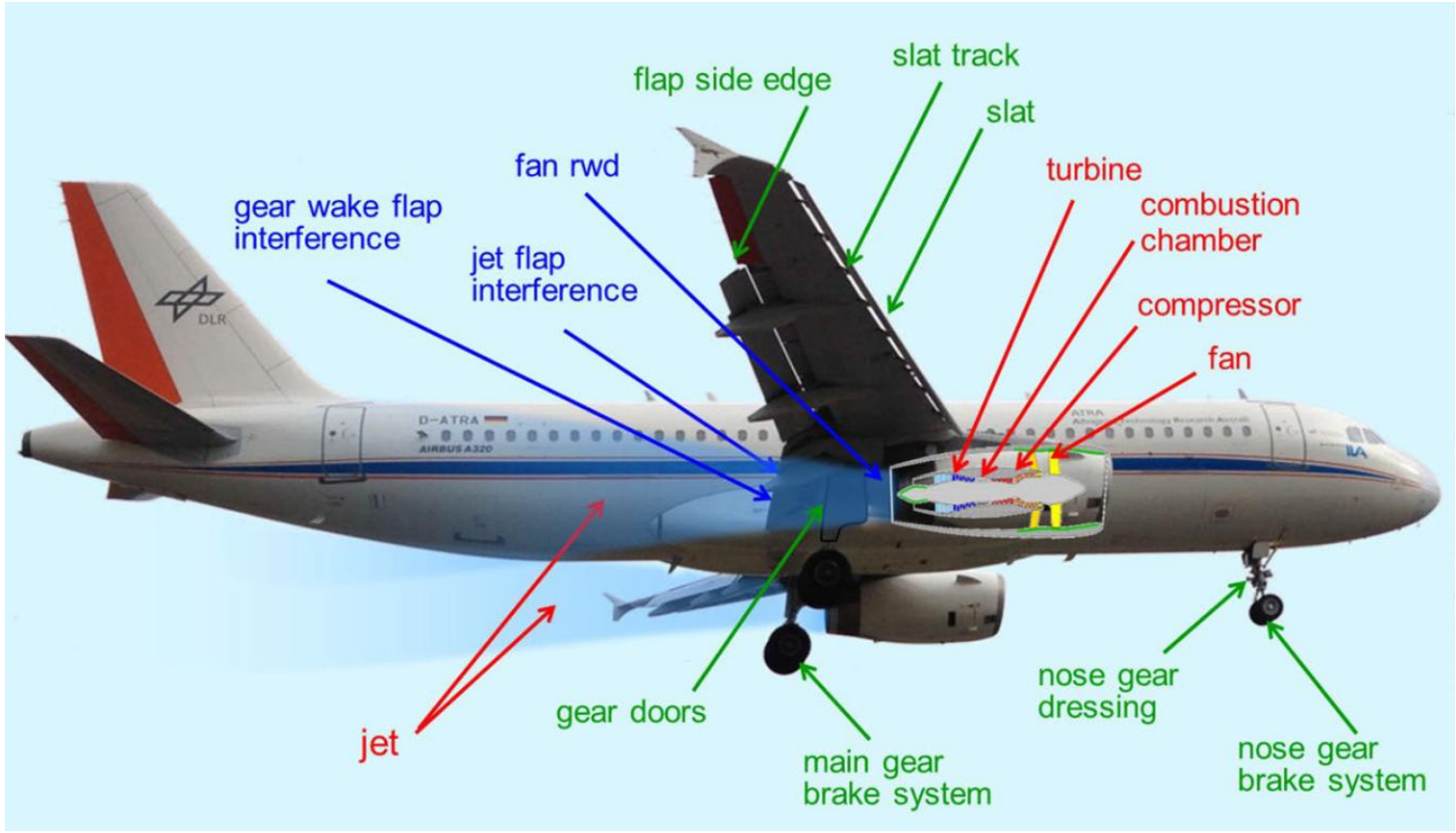
- Increase in stringency of **7 EPNdB (cumulative) relative to the Chapter 4** cumulative levels
- Applicability **to new types submitted for certification on or after 31 December 2017** (31 December 2020 for aircraft with MTOW < 55 tons)
- Supplementary condition of not less than 1.0 dB below Chapter 3 limits at each certification point
- Change in the noise limits applicable to subsonic jet aeroplanes with MTOW < 8,618 kg



Chapter	Applicable Year	Cum Margin (EPNdB)
2	1972	Ch. 3 ~-16
3	1978	0 (Ref)
4	2006	Ch. 3 +10
14	2017 & 2020	Ch. 3 +17 (Ch. 4 +7)

Source: ICAO

Sources of sound emissions at the aircraft

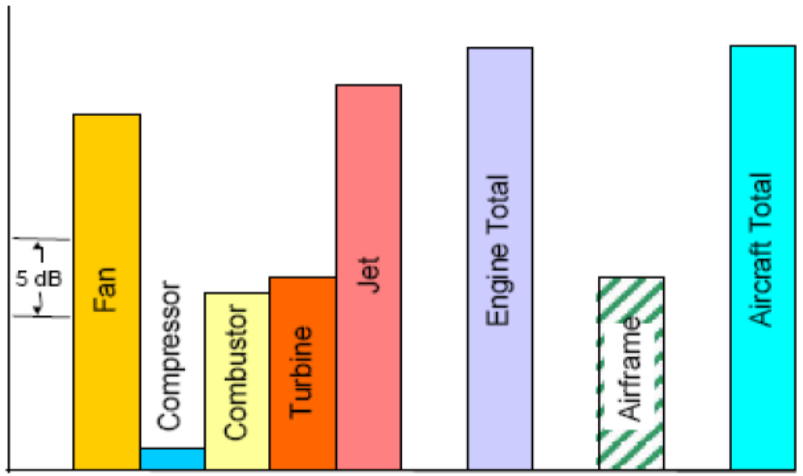


- During takeoff, the **engine-related noise** is predominant.
- During landing, the **aircraft structure-related noise** plays the most important role.

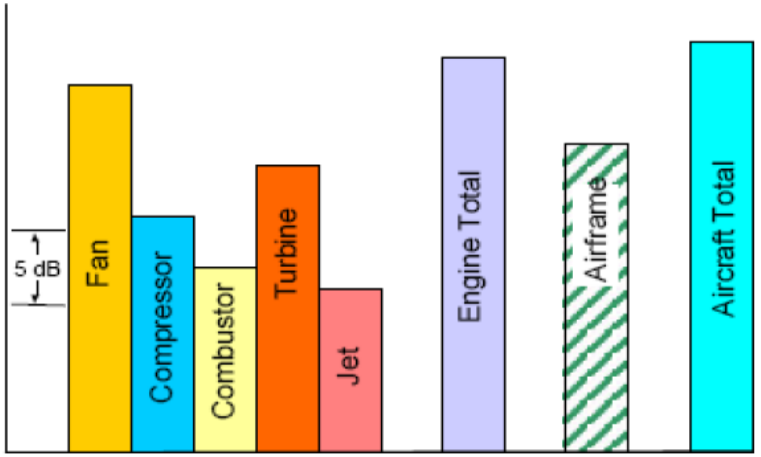
Sources:

DLR
Bertsch et al., 2019

Source noise component contributions



Typical Departure
Noise Distribution



Typical Arrival
Noise Distribution

Source: ICCAIA

Engine sources of noise

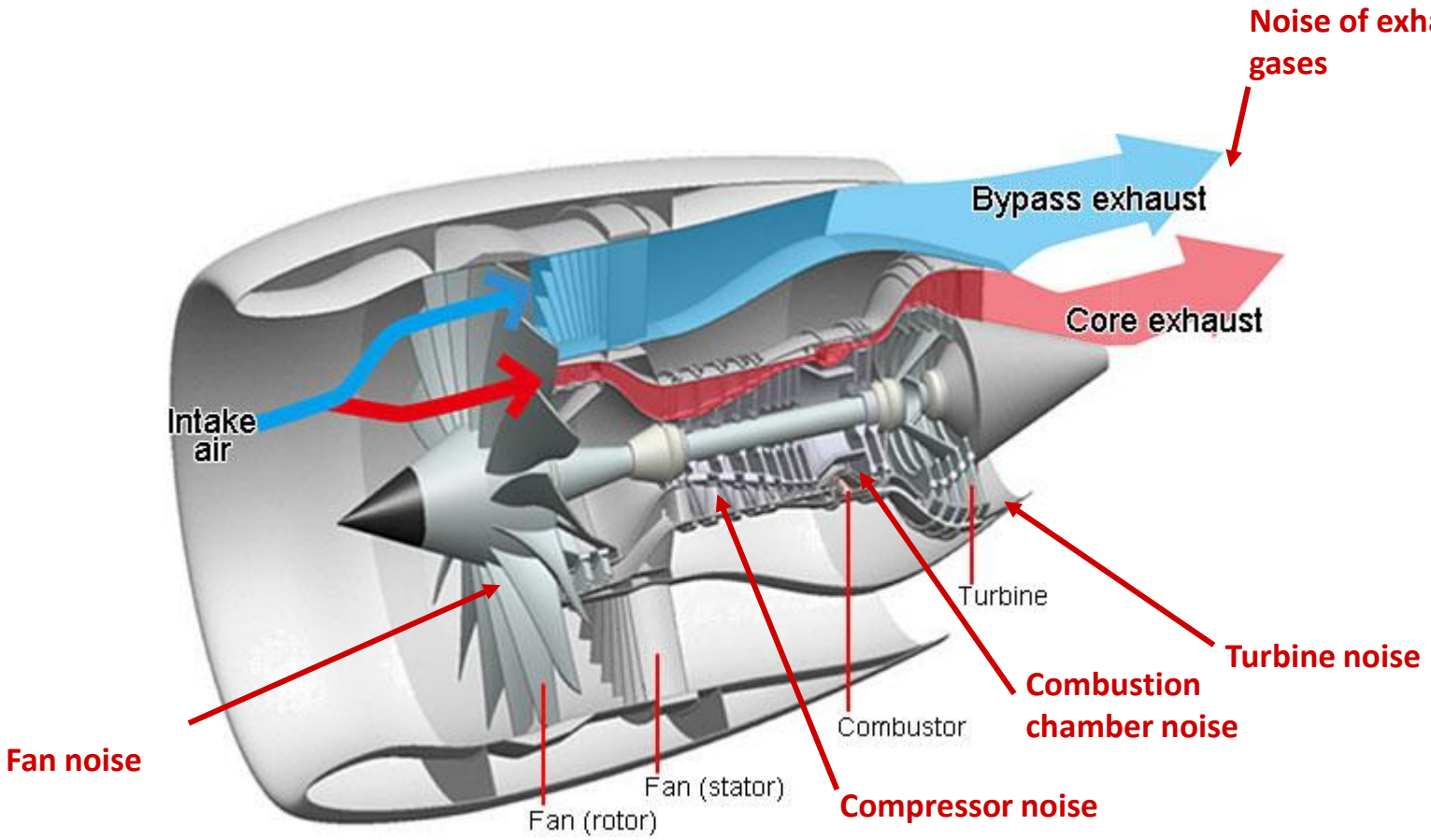
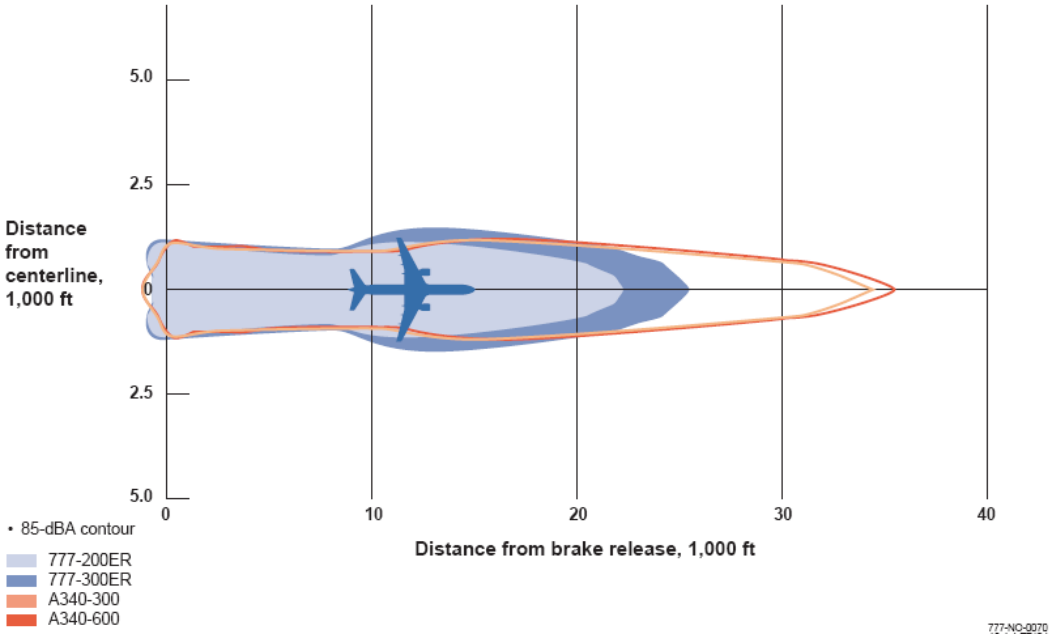


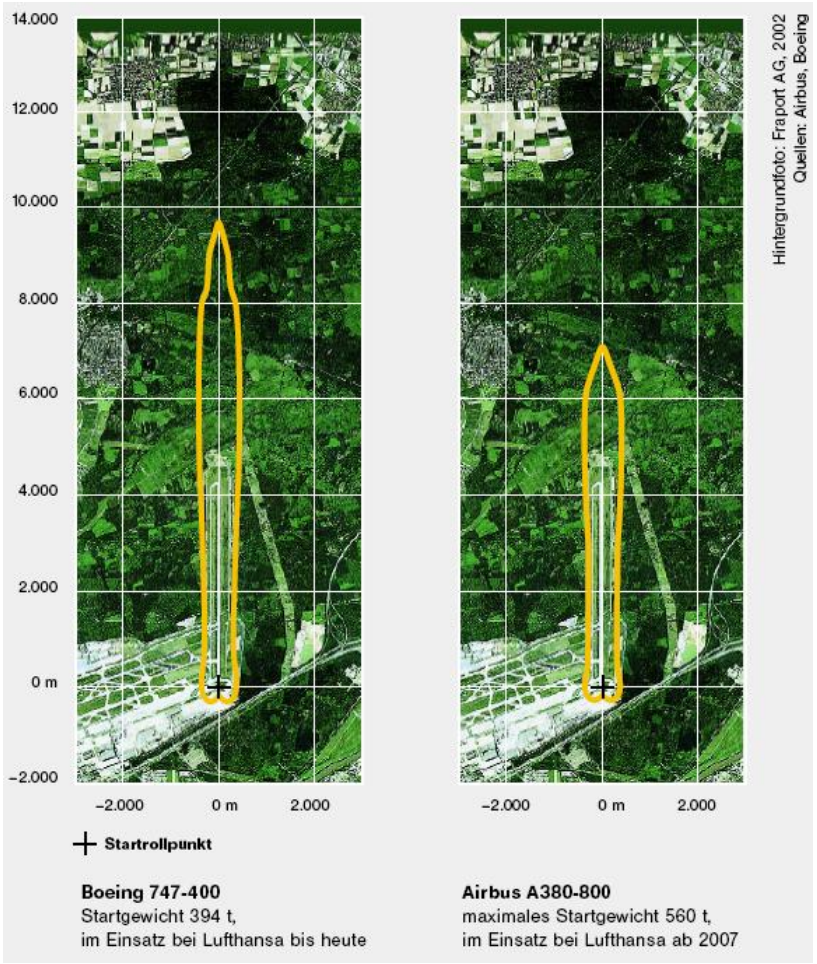
Image Source: JAXA

Noise emissions: Noise carpets



Boeing 777 vs. Airbus A340: Takeoff 85 dB(A) contours

- An aircraft does not emit noise into all directions equally.
- Every aircraft features a characteristic noise contour (lines of equal dB-values)



Source: Lufthansa

Boeing 747-400 vs. Airbus A380-800: Takeoff L_{Amax} 85 dB(A) contours

Noise mitigation measures for the aircraft structure

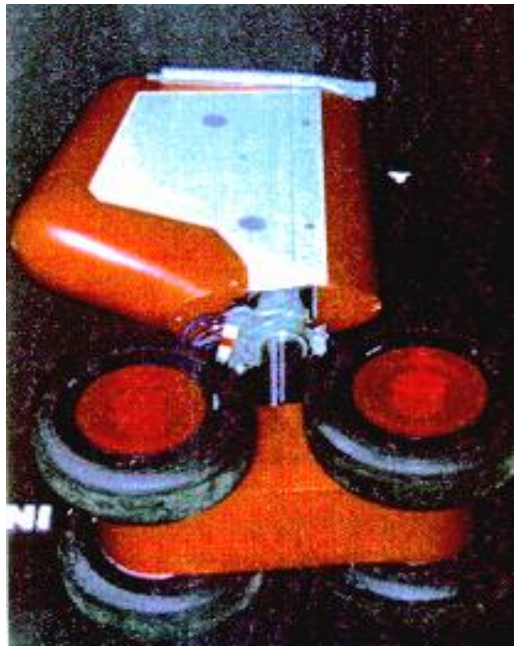
Short-term measures: experimental methods (e.g., wind tunnel testing) aimed at optimizing the noise characteristics of the landing gear and the high-lift devices.

Long-term measures:

- Calculation of noise emissions and local distribution through CAA (Computational Aeroacoustics)
- Noise-focused optimization through numerical simulation of various configurations.



Left: Acoustic localization of noise sources through wind tunnel testing



Right: Noise-related optimization of the landing gear

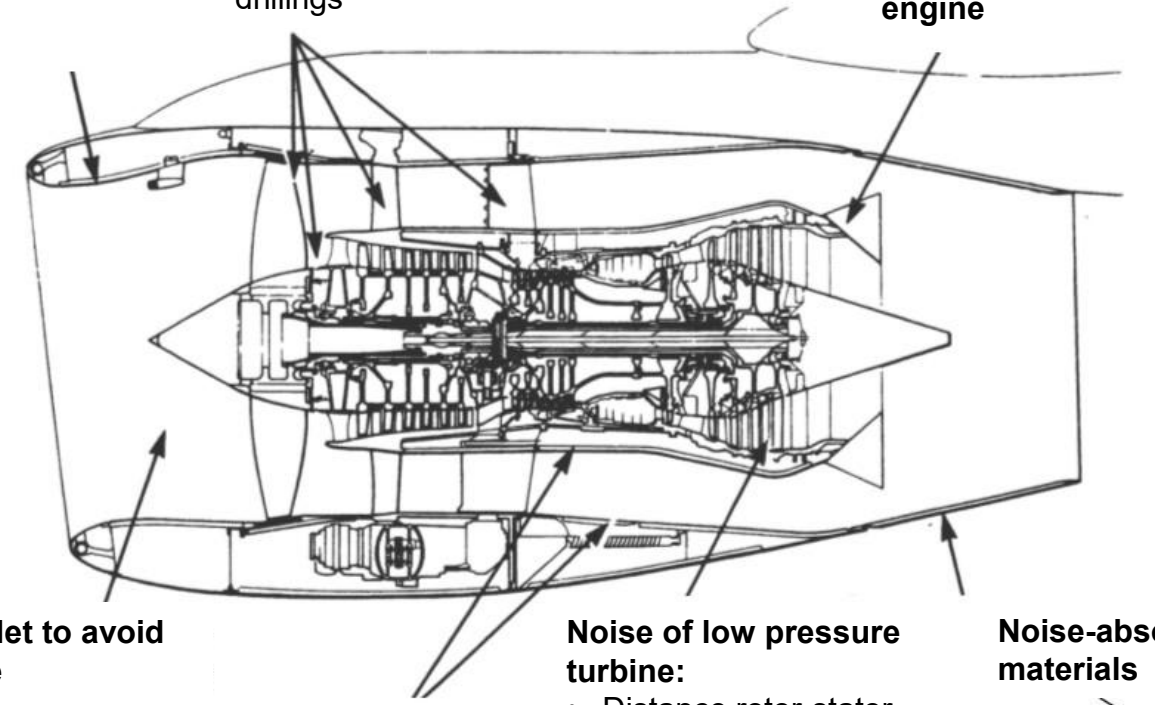
Source: DLR

Noise mitigation measures for modern turbofan engines

Fan noise:
- Distance rotor – stator
- Number of the rotor and stator blades
- Little number of maintenance drillings

Noise-absorbing material

Mixing the bypass and the core airflows within the engine

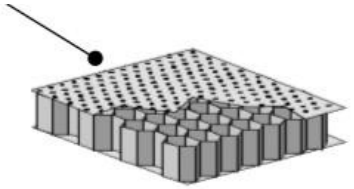


Smooth inlet to avoid turbulence

Noise-absorbing materials

Noise of low pressure turbine:
- Distance rotor-stator
- Number of rotor and stator blades

Noise-absorbing materials



Noise restrictions at airports - Overview

Operational noise restrictions:

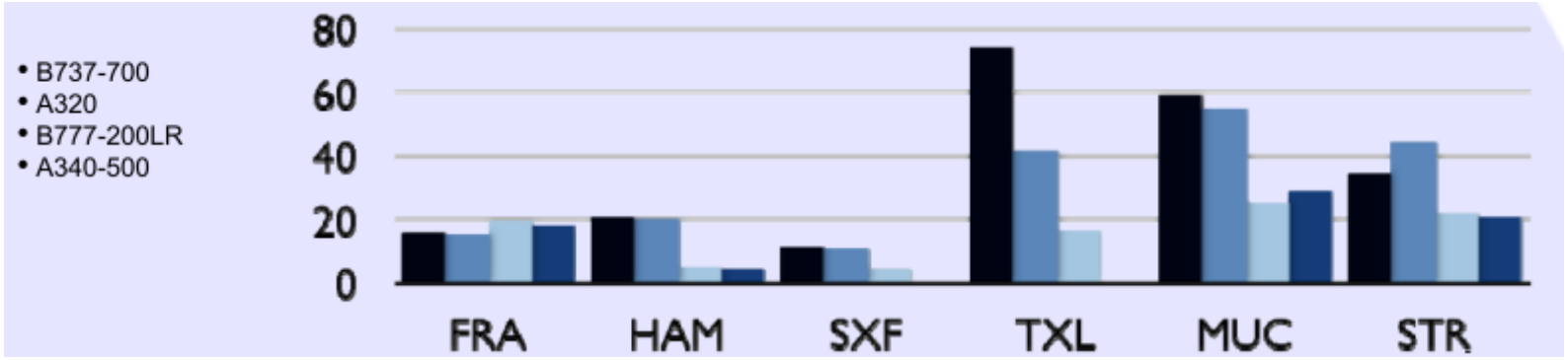
- **Night curfews:**
  - Predominant at European airports
  - Application cases: only one runway closed / operating hours according to maximum noise levels / general night curfew
- **Limitation of noise level:**
  - Flying ban when a certain noise level is exceeded
  - Penalty charges when a certain noise level is exceeded
- **Quota regulations:**

  Maximum number of flights per airline or per time interval and slot allocation according to aircraft noise characteristics
- **Noise charges:**

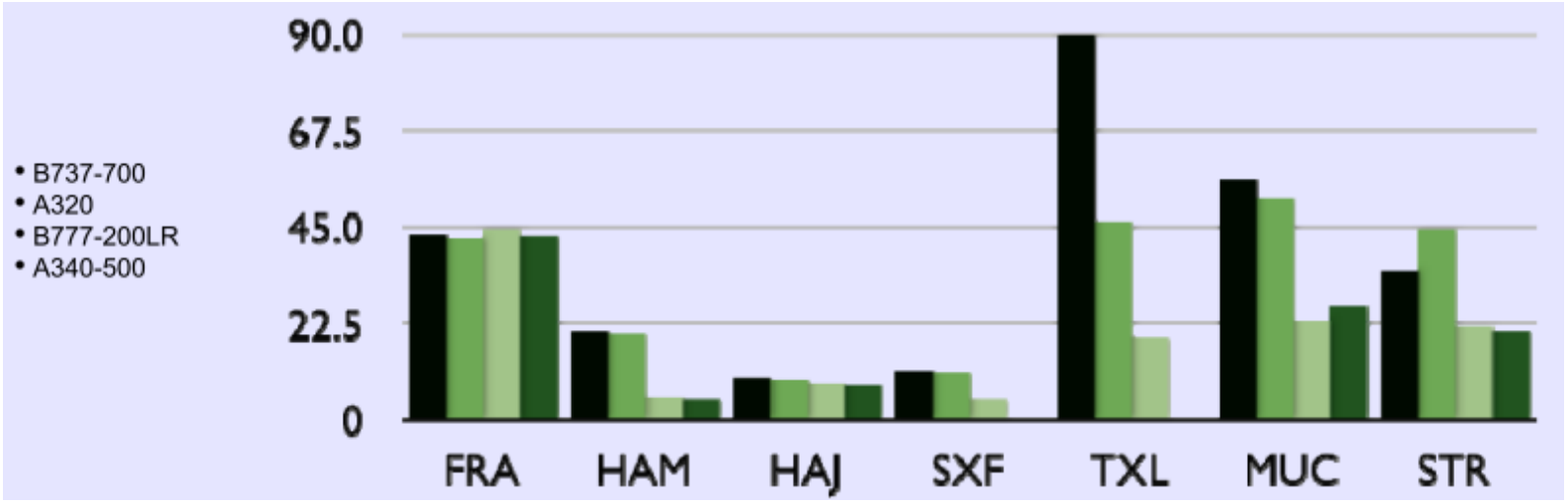
  Factors involved are the noise level, the time of departure and arrival, and the season
- **Noise-reducing flight procedures / Noise abatement procedures:**
  - Vertical and horizontal noise abatement procedures
  - Direct noise reduction through ‘Thrust Cutback’
  - Steep departing and approach procedures help reduce ground noise considerably

Noise charges at German airports (2010)

Proportion of noise fee in relation to total landing fee (**day**) [%]

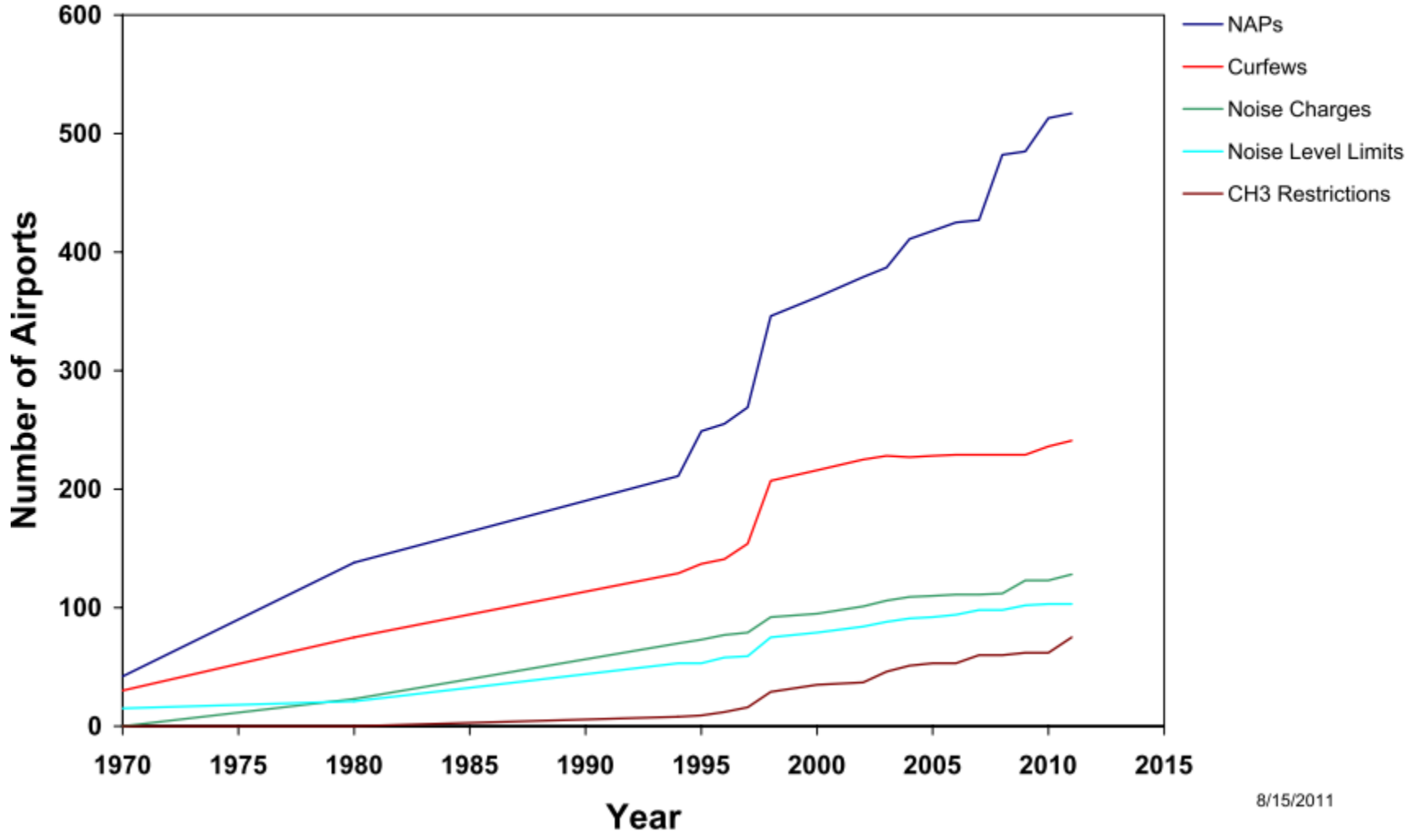


Proportion of noise fee in relation to total landing fee (**night**) [%]



Source: DLR

Growth in airport noise restrictions



Source: Boeing

Excursus: Noise simulation in the airport vicinity (1)

Software: Aviation Environmental Design Tool (AEDT) – basic principles

- AEDT by FAA models aircraft performance in order to calculate noise, fuel burn and emissions such as particular matters or NO_x.
- Noise simulation within AEDT: Intended for calculation of cumulated noise impact in a certain period of time and not for a single flight event (limited accuracy).

Major simulation input data consist of:

- Acoustic data as a function of engine power and distance to the point of immision, so- called Noise-Power-Distance (NPD) Data which represent
  noise propagation. Distance data lie between 200 and 25.000 ft.
  - ➢ Separate data for different noise metrics (e.g. SEL, EPNL)
  - ➢ Separate data for either approach or departure
- Flight plan data, with focus on occurrence (day, night,
  evening) and used departure/arrival route.

Excursus: Noise-Simulation in the airport vicinity (2)

Software: Aviation Environmental Design Tool (AEDT) – simplified noise calculation sequence

1. Iterative process of calculating noise for each single aircraft operations
- Aircraft power usage for every flightpath segment and the slant distance between the aircraft's trajectory and point of immission is calculated.
- Points of immision are implemented as a grid of points on the ground with a chosen resolution, a so-called receptor grid.
- Noise levels are being calculated based on the NPD-tables corresponding to the operational mode of the segment (approach or departure). In order to get noise levels between tabulated values, linear approximation is used for thrust values and logarithmic approximation on distance values.

2. Accumulation of single-event noise and computation of chosen cumulated noise metric.

Conclusion:

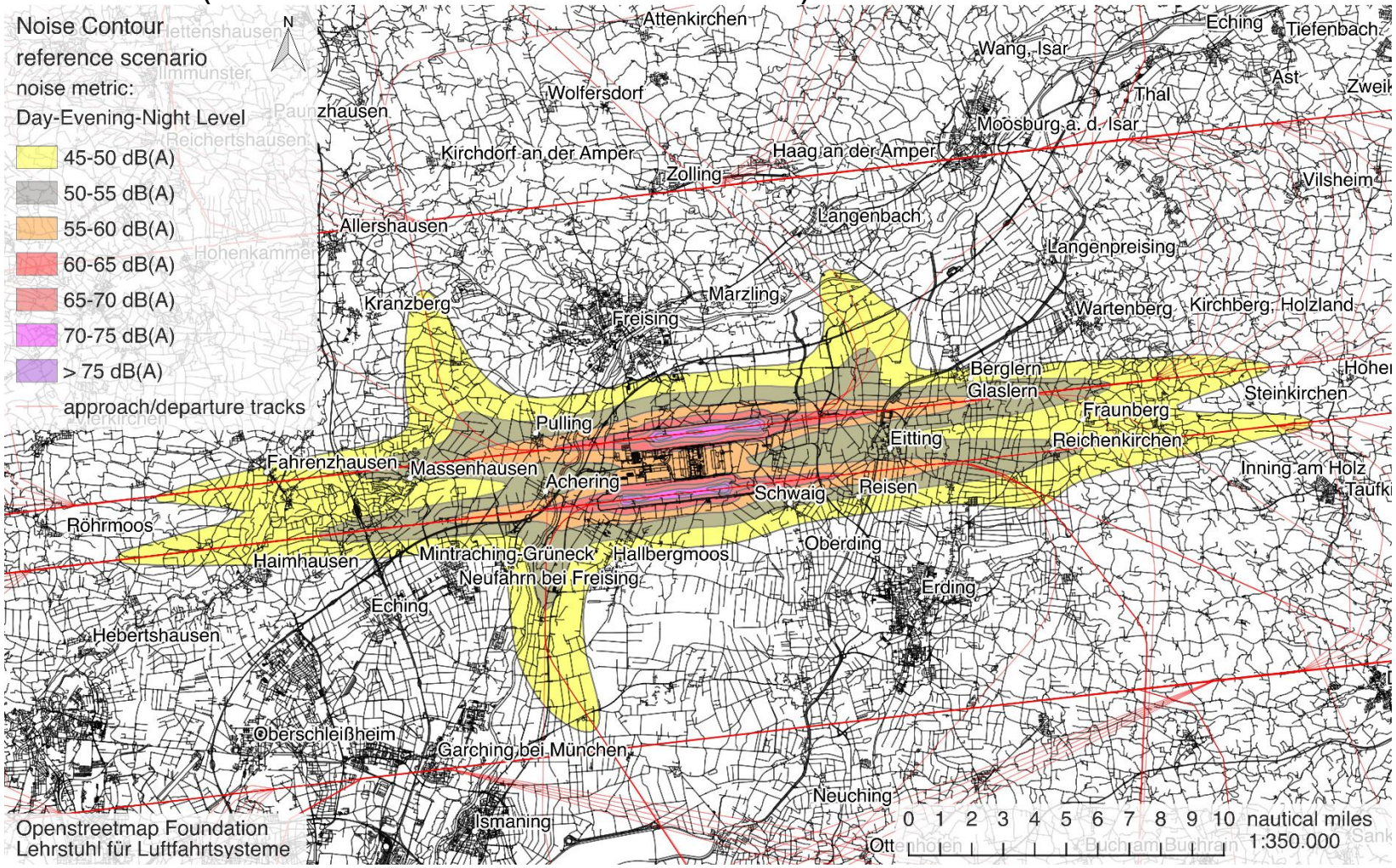
Evaluation of airport noise impact caused by airport infrastructure extension, changes in fleets, air movement growth, optimized ground tracks or due to improved noise abatement procedures.

Next slides: Exemplary study of airport noise contours

- Munich Airport (EDDM)
- Two scenarios: Reference scenario vs. growth scenario (16 years later, exemplary annual growth rate in aircraft movements of 3 percent)

Excursus: Noise-Simulation in the airport vicinity (3)

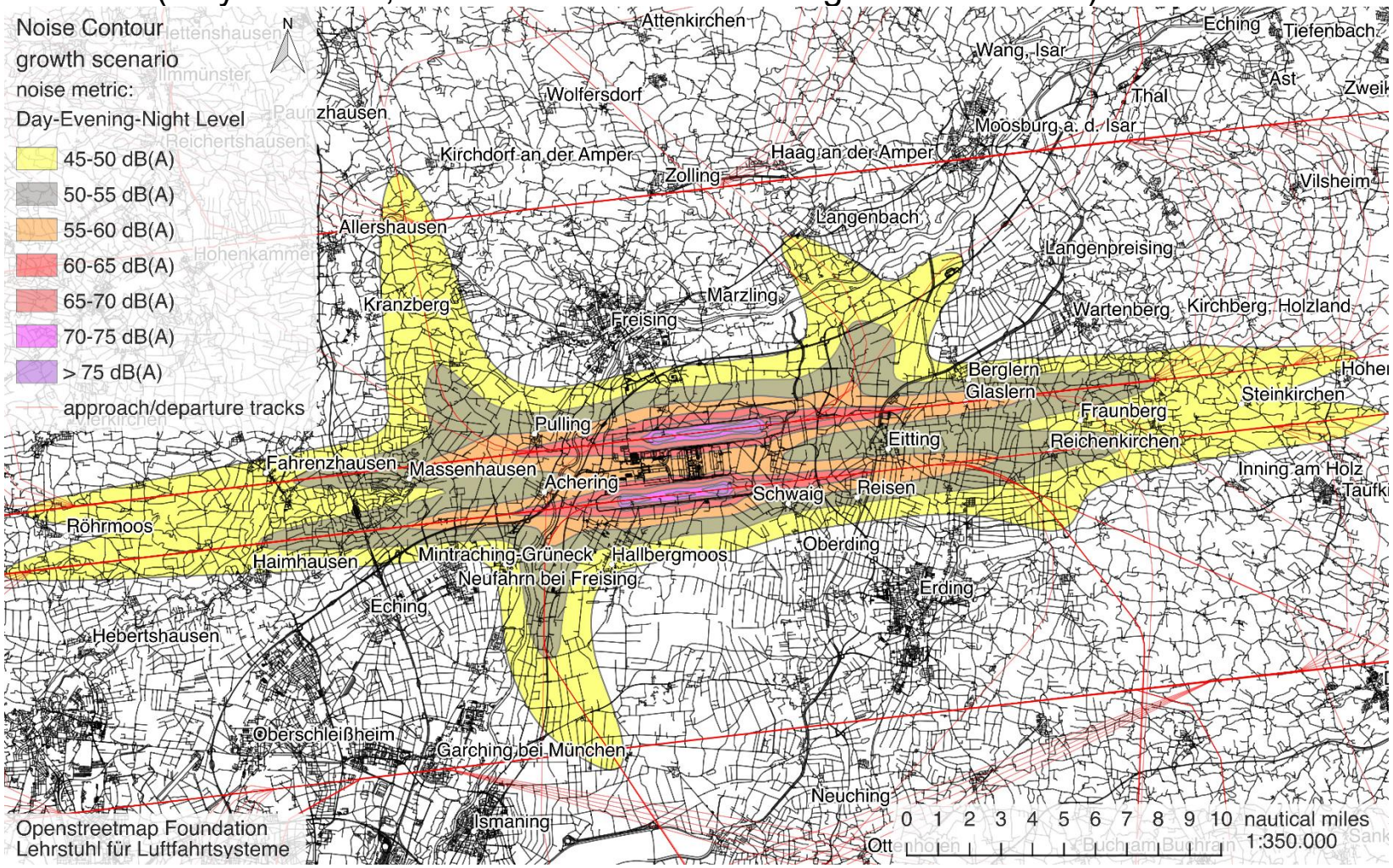
AEDT – reference scenario (maximum of ca. 60 movements/hour)



Disclaimer: The above figure is based on non-realistic assumptions concerning the number of aircraft movements and is thus intended to be for illustration purposes only.

Excursus: Noise-Simulation in the airport vicinity (4)

AEDT – growth scenario (16 years later, annual aircraft movement growth rate: 3 %)



Disclaimer: The above figure is based on non-realistic assumptions concerning the number of aircraft movements and is thus intended to be for illustration purposes only.

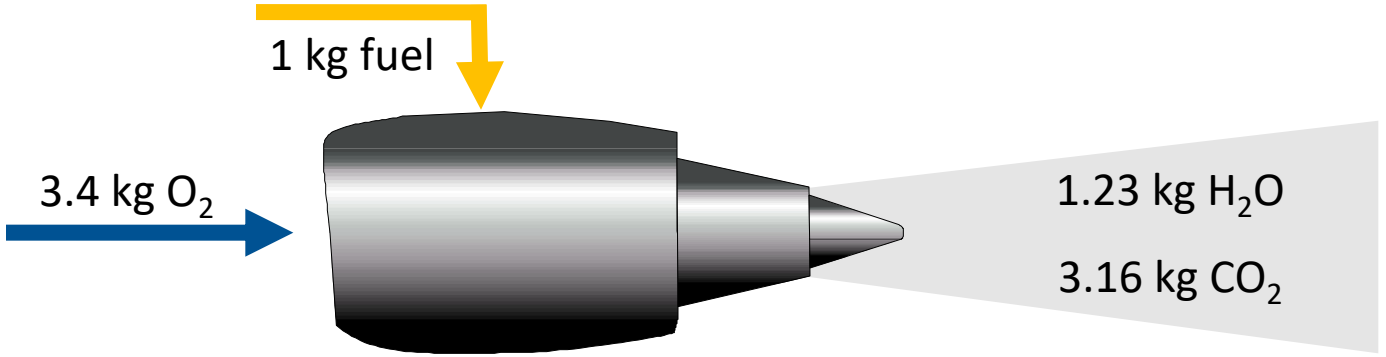
7. Environmental Impact of Aviation

7.2 Aircraft Exhaust Emissions



Exhaust emissions produced when burning jet fuel

For 1 kg of jet fuel (kerosene), approx. 4.4 kg of exhaust emissions are produced (stoichiometric combustion)



Variable quantities of exhaust emissions:

Nitrogen oxides NOx	~ 4 - 40g
Sulphur oxides SOx	~ 0.6 - 1g
Soot	~ 0.01 - 0.03g
Carbon monoxide CO	
Unburned Hydrocarbons UHC	

Traces of OH, methane, lubricating oils, water vapor (contrails)

→ generates / destroys ozone O_3

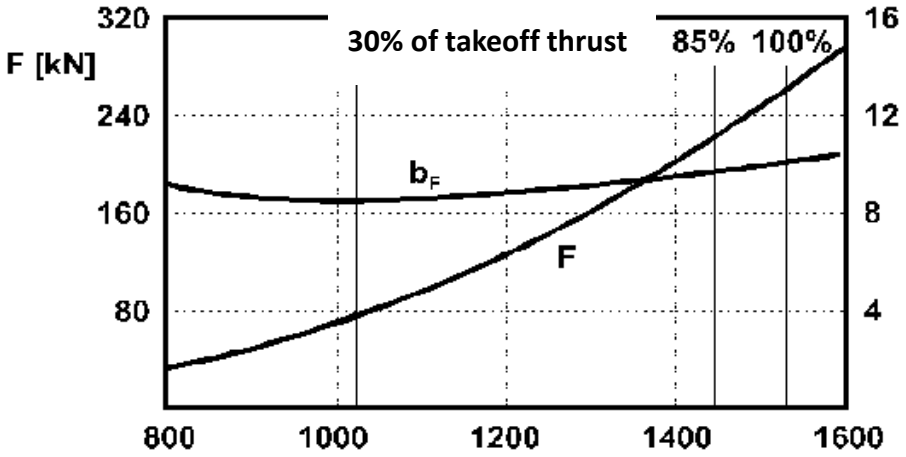
→ destroys methane CH_4

→ ozone and methane are greenhouse gases

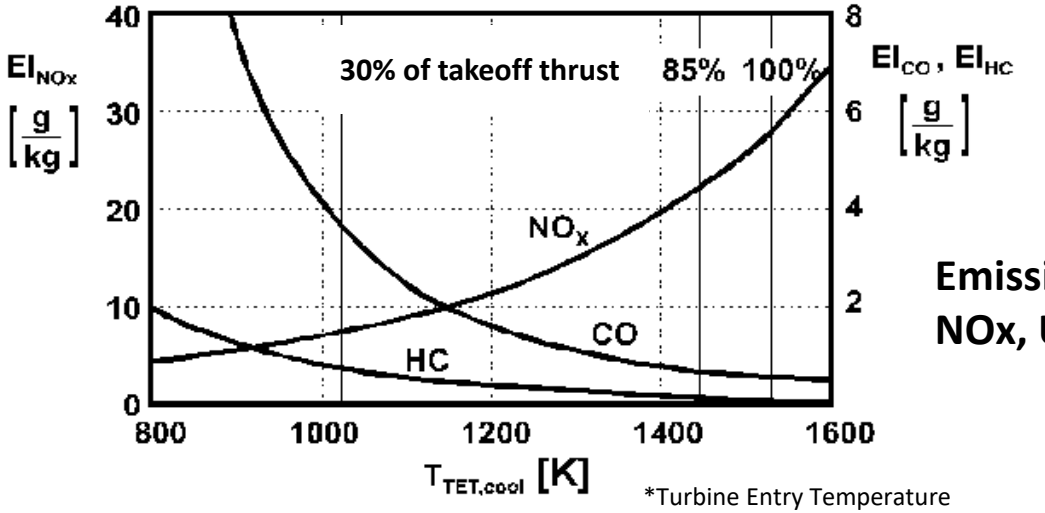
The **emission index** is used to indicate the quantity of a specific pollutant per kilogram of jet fuel burned.

Emission Index EI [g/kg fuel]

Engine emissions as a function of the TET*



Specific Fuel Consumption



Emissions Indices for NOx, UHC, and CO

*Turbine Entry Temperature

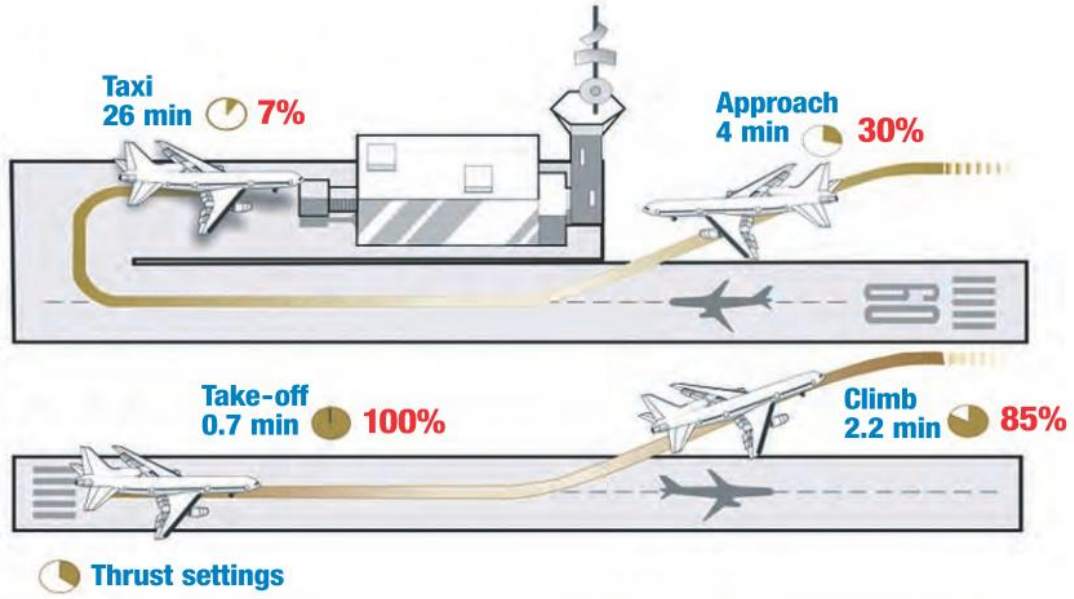
Source: DGLR-Jahrbuch 1994

Important emission producers at an airport



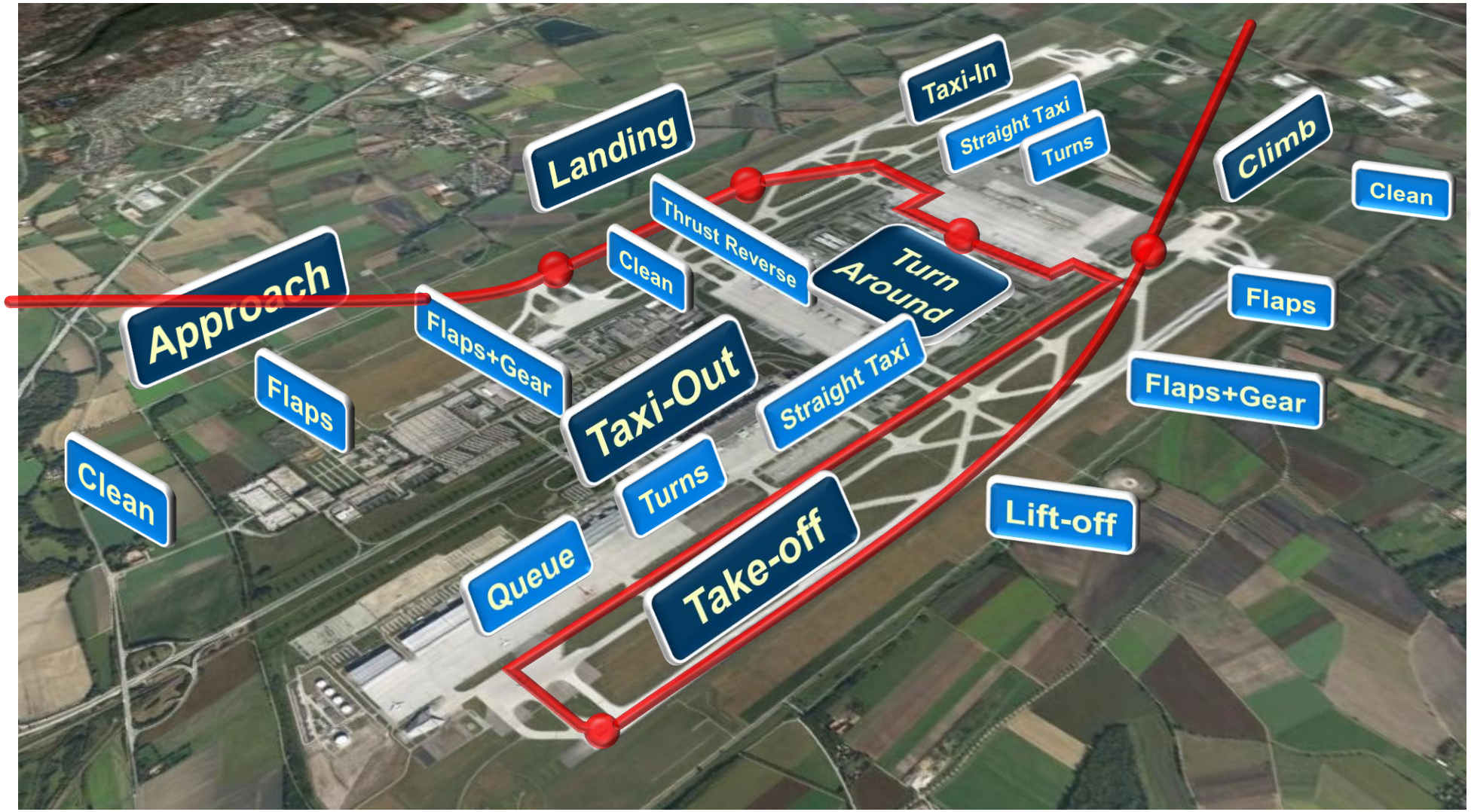
The ICAO Landing & Takeoff (LTO) Cycle

- The LTO cycle was defined in the 1970s. This reference cycle captures aircraft emissions from MSL to an altitude of up to 3,000 ft. All emissions that are produced below this altitude contribute to the **local air quality** in the airport vicinity.
- When a new engine is being certified, all **emissions measurements are referenced to the LTO cycle at MSL and the International Standard Atmosphere** (ISA).
- The LTO cycle is applicable for **simple estimation calculations**. It is not appropriate for detailed emissions studies.
- The LTO cycle is a **model cycle and therefore does not capture** actual pilot behavior, flight profiles, aircraft performance, and atmospheric conditions.



Source: ICAO

Detailed approach to local emissions calculation



Pollution concentration at Munich Airport

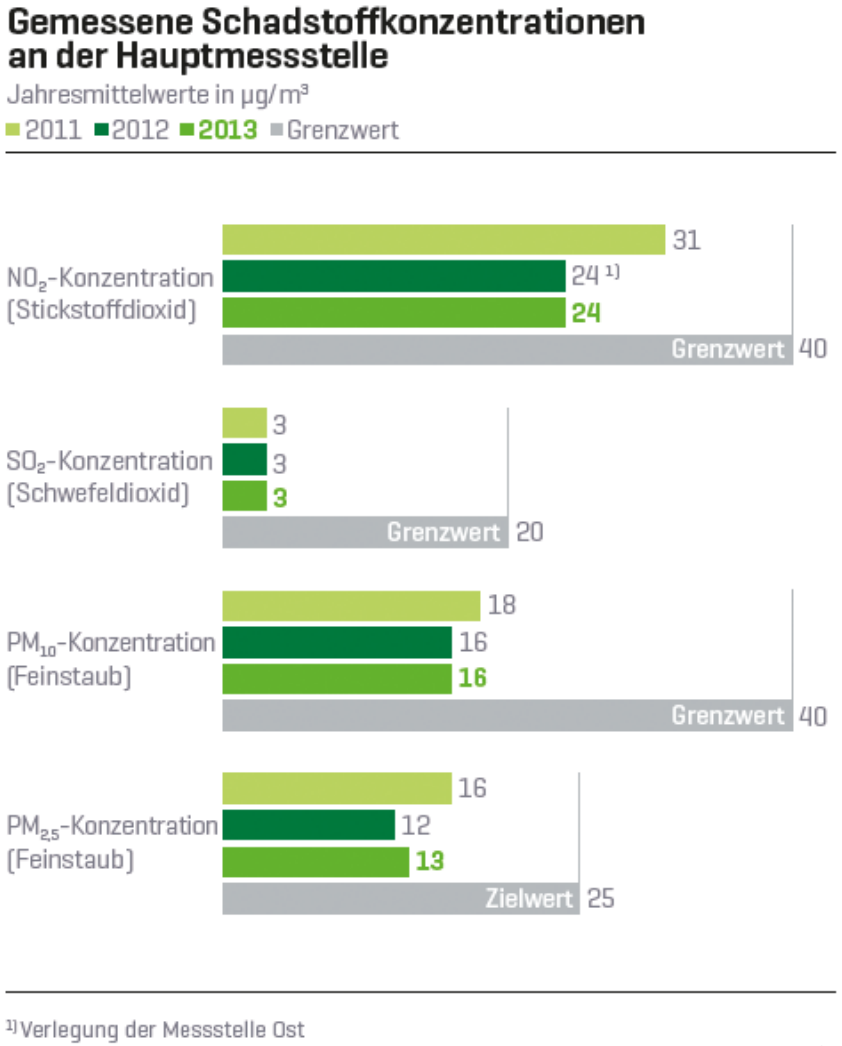
Pollution concentrations measured in 2011, 2012, and 2013

NO_2 Concentration

SO_2 Concentration

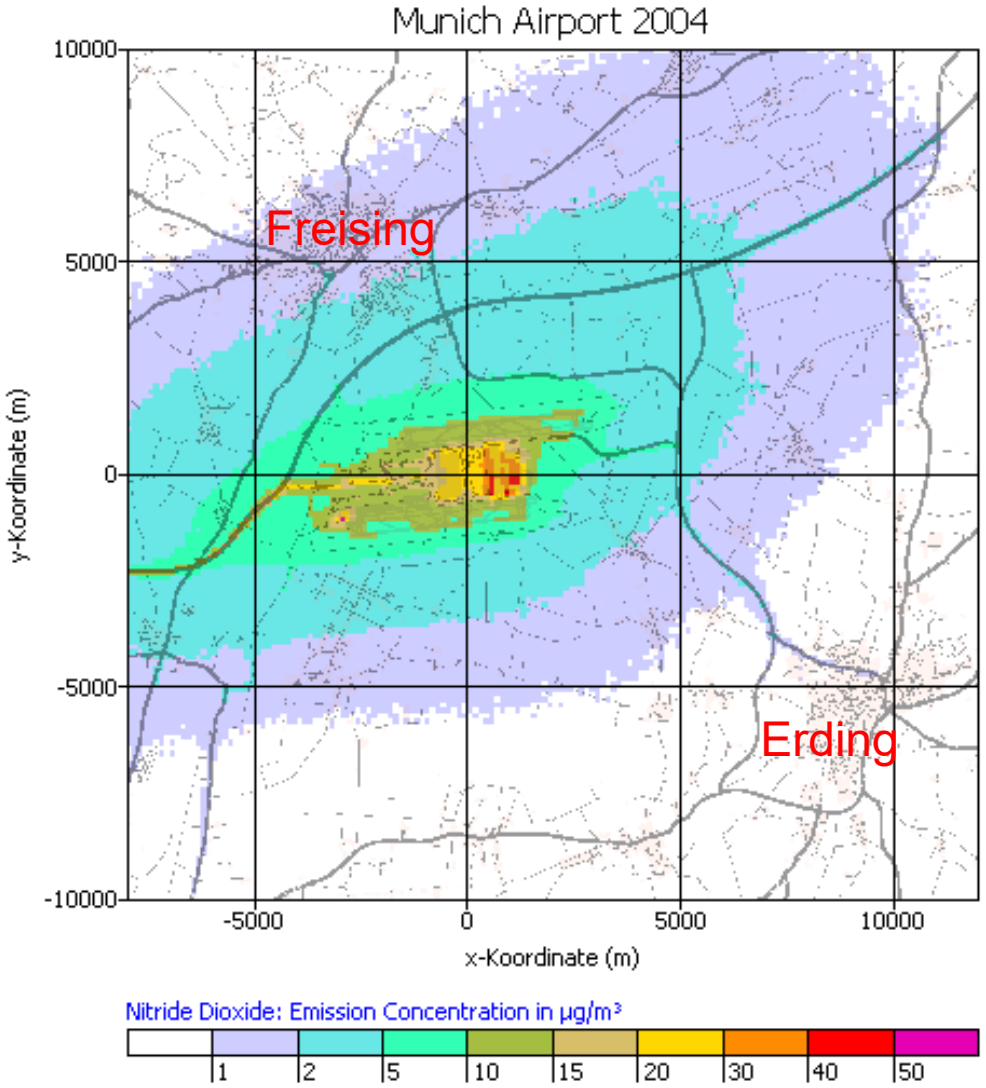
PM_{10} Concentration

$PM_{2.5}$ Concentration



Source: Munich Airport

NO_2 pollution at Munich Airport (1)

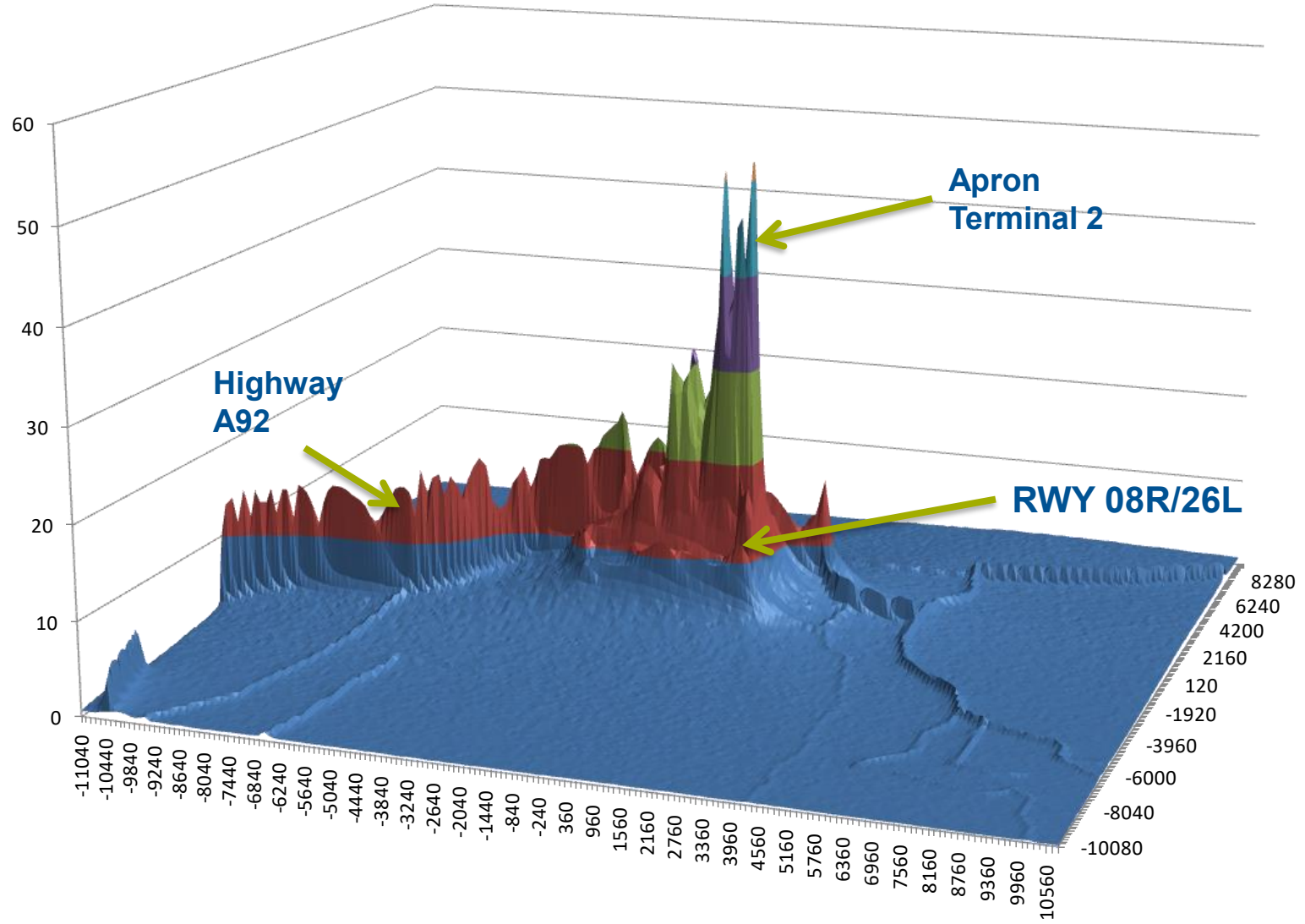


From 2010, a yearly average value of 40 µg/m³ must not be exceeded at German airports.

Outside the airport vicinity area, Munich Airport contributes roughly 5 – 8µg/m³ of NO_2 on yearly average. This represents 12.5 – 20% of the legal limiting value.

Source: Munich Airport

NO_2 pollution at Munich Airport (2)



Source: Munich Airport

Health effects of air pollutants

Ozone: Lung function impairment, effects on exercise performance, increased airway responsiveness, increased to respiratory infection, increased hospital admissions and emergency room visits, and pulmonary inflammation, lung structure damage.

Carbon Monoxide: Cardiovascular effects, especially in those persons with heart conditions (e.g., decreased time to onset of exercise-induced angina).

Nitrogen Oxides: Lung irritation and lower resistance to respiratory infections

Particulate Matter: Premature mortality, aggravation of respiratory and cardiovascular disease, changes in lung function and increased respiratory symptoms, changes to lung tissues and structure, and altered respiratory defense mechanisms.

Volatile Organic Compounds: Eye and respiratory tract irritation, headaches, dizziness, visual disorders, and memory impairment.

Source: United States Environmental Protection Agency (1999)

Environmental effects of air pollutants

Ozone: Crop damage, damage to trees and decreased resistance to disease for both crops and other plants.

Carbon Monoxide: Similar health effects on animals as on humans.

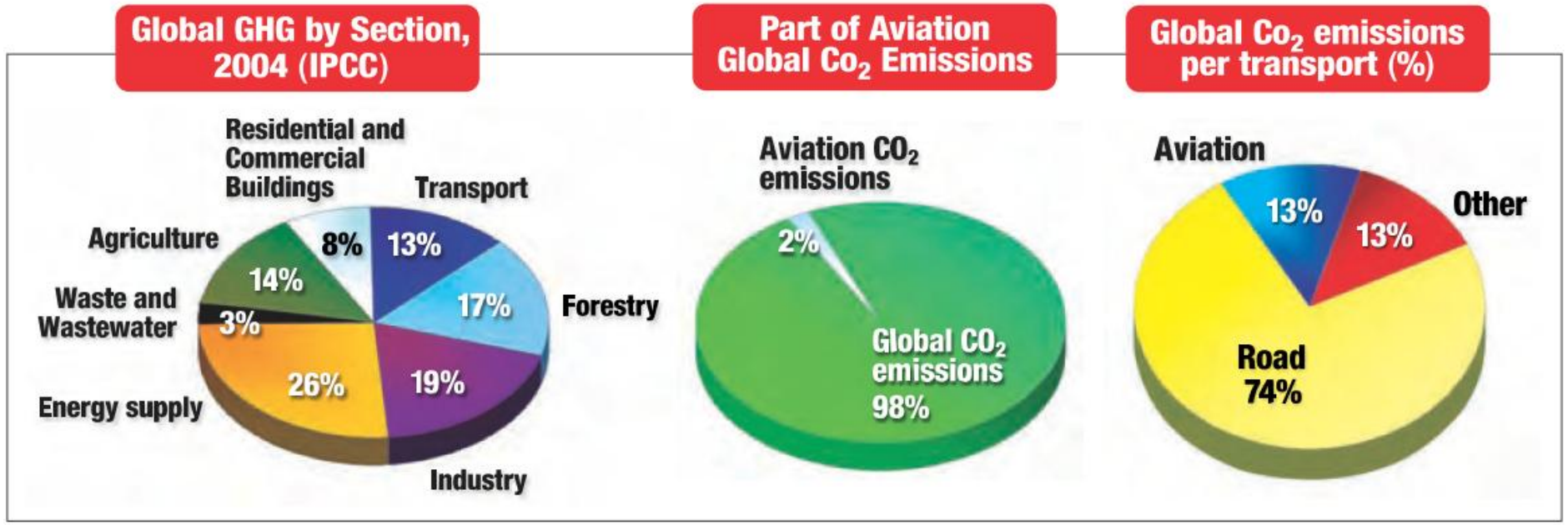
Nitrogen Oxides: Acid rain, visibility degradation, particle formation, contribution towards ozone formation.

Particulate Matter: Visibility degradation and monument and building soiling, safety effects for aircraft from reduced visibility.

Volatile Organic Compounds: Contribution towards ozone formation, odors, and some direct effect on buildings and plants.

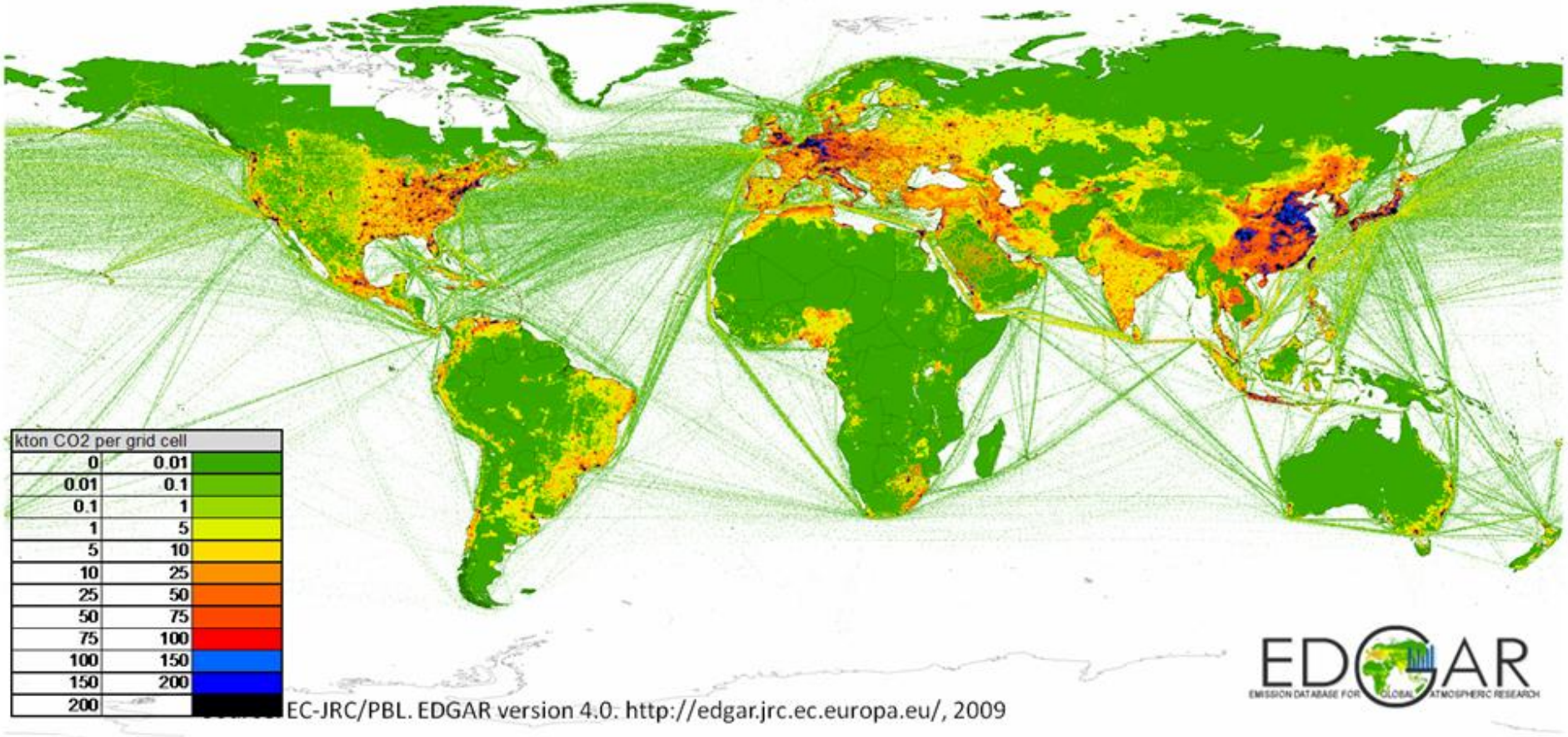
Source: United States Environmental Protection Agency (1999)

Aviation's contribution to global CO2 emissions



Source: ICAO (2010)

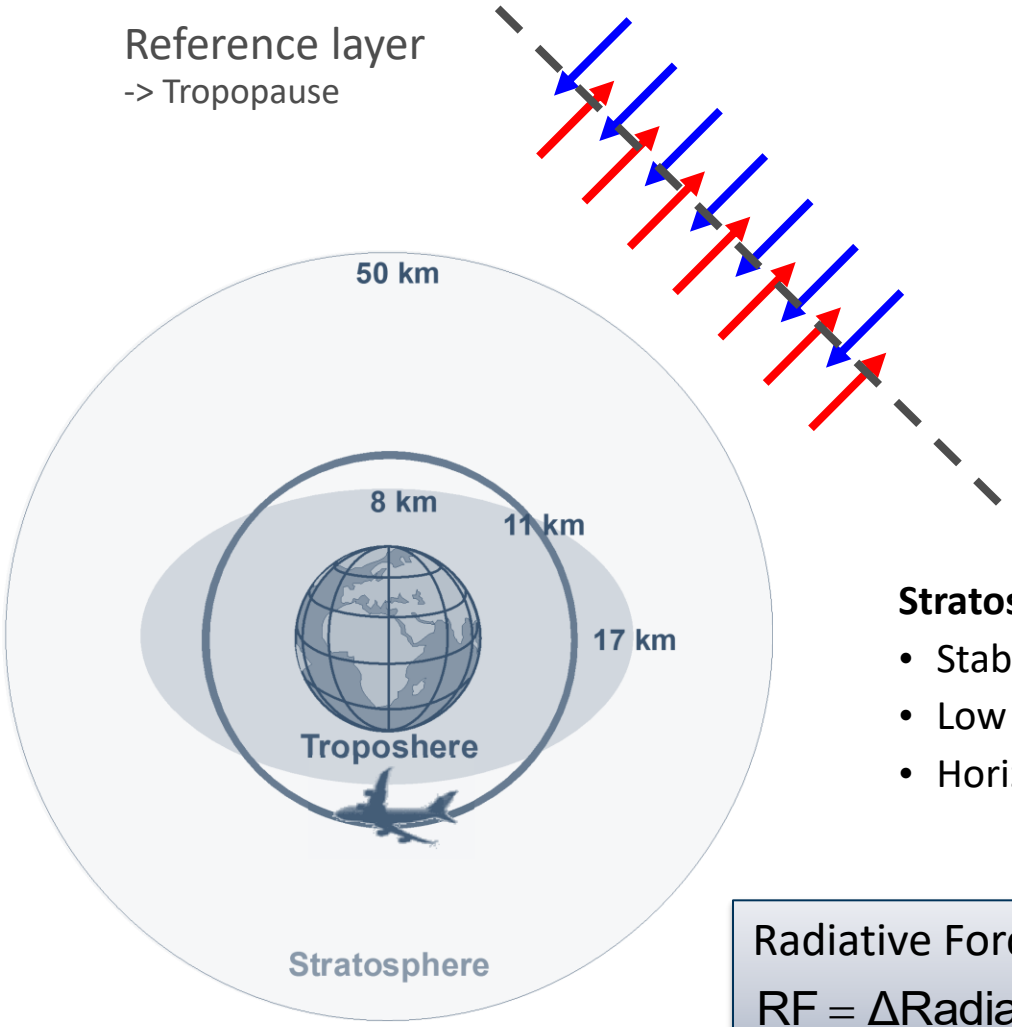
Global carbon dioxide emissions in 2005 (including aviation)

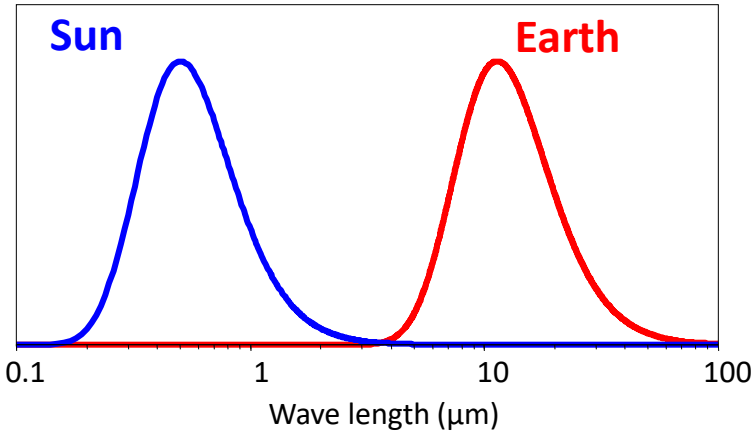


Source: EU Joint Research Center

Effect of greenhouse gases

Reference layer
-> Tropopause





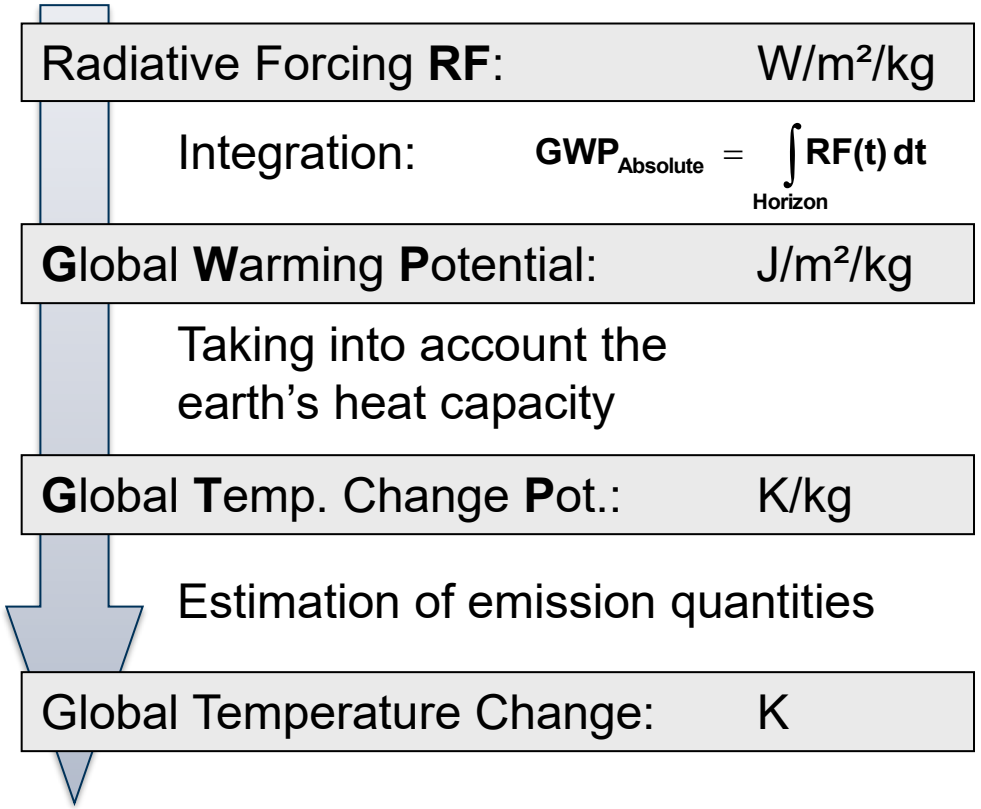
Stratosphere
- Stable layers
- Low humidity
- Horizontal winds

Troposphere
- “Weather layer”
- High humidity
- Horizontal and vertical winds

Radiative Forcing (RF)

$$RF = \Delta Radiation = Radiation_{Sun} - Radiation_{Earth}$$

Climate-related metrics based on radiative forcing



(Metrics of the Kyoto Protocol)

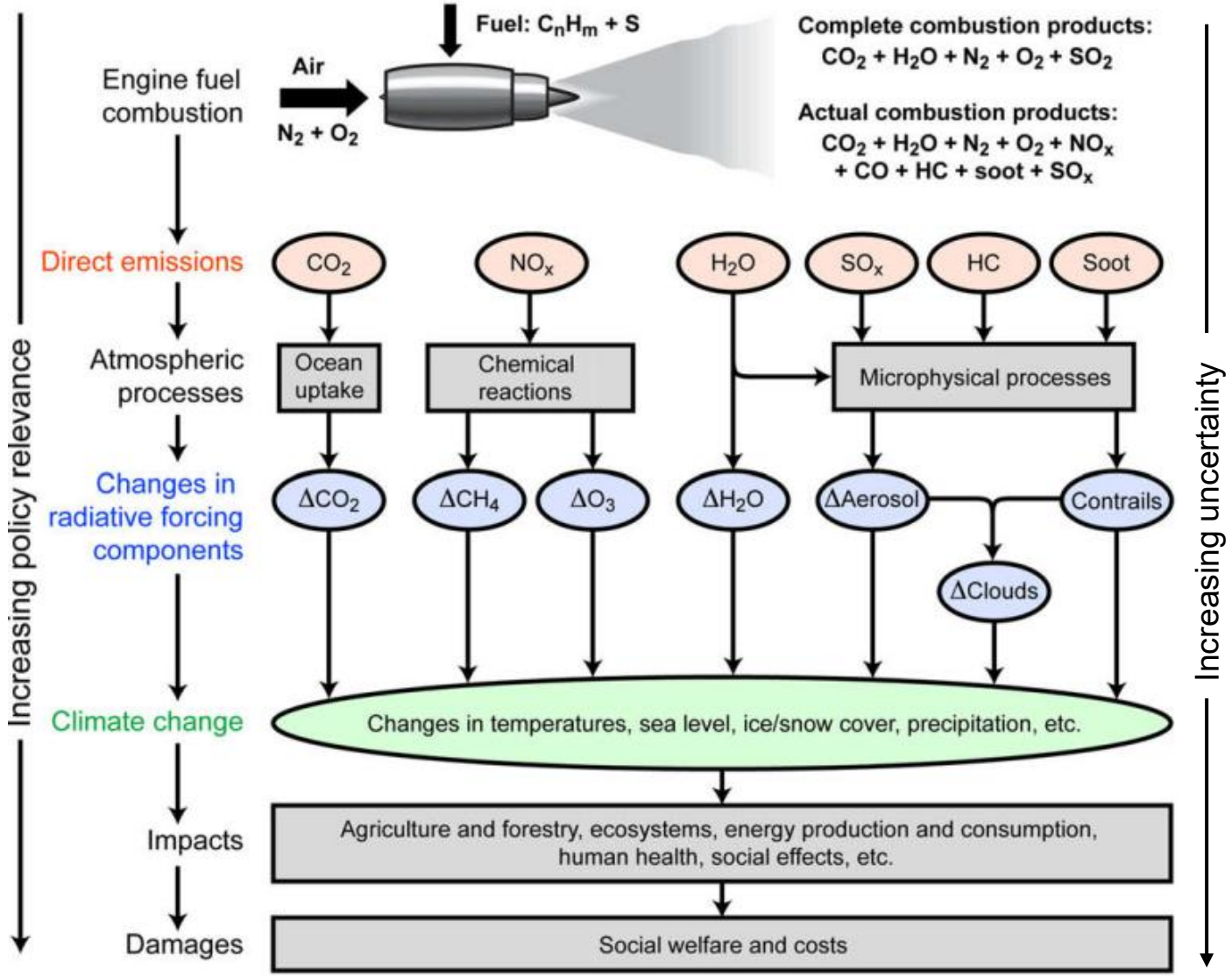
Empirical model for durable gases (aviation: only CO_2)

$$\mathbf{\Delta Temp} = \lambda \cdot \mathbf{RF}$$

Disadvantage: Value of λ depending on the model and on the gas!

Sources:
Shine et al., 2005
Berntsen et al., 2005
Schumann et al., 2002

Aircraft emissions and climate change

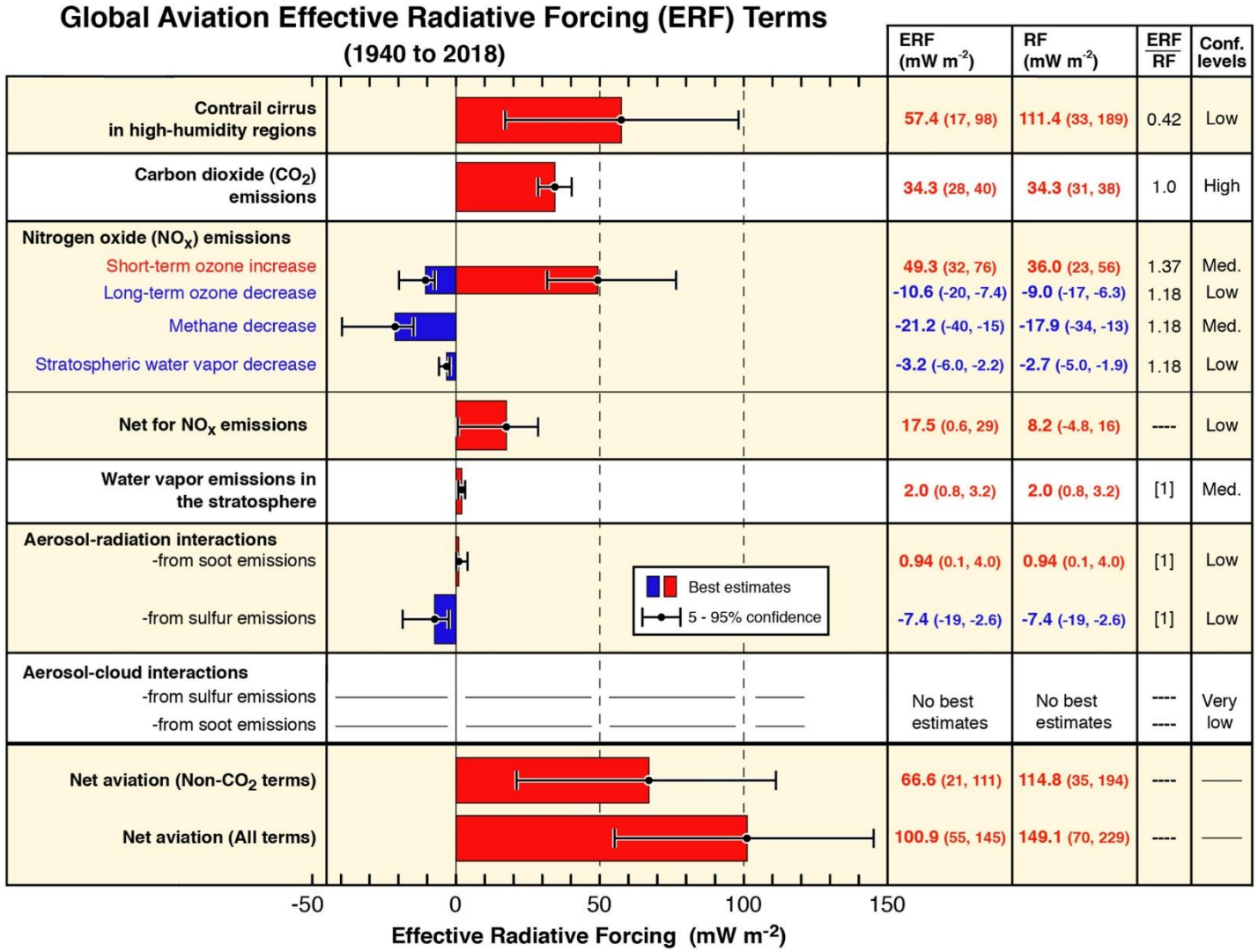


Source:

Lee et al. (2009)

Fuglestvedt et al. (2003)

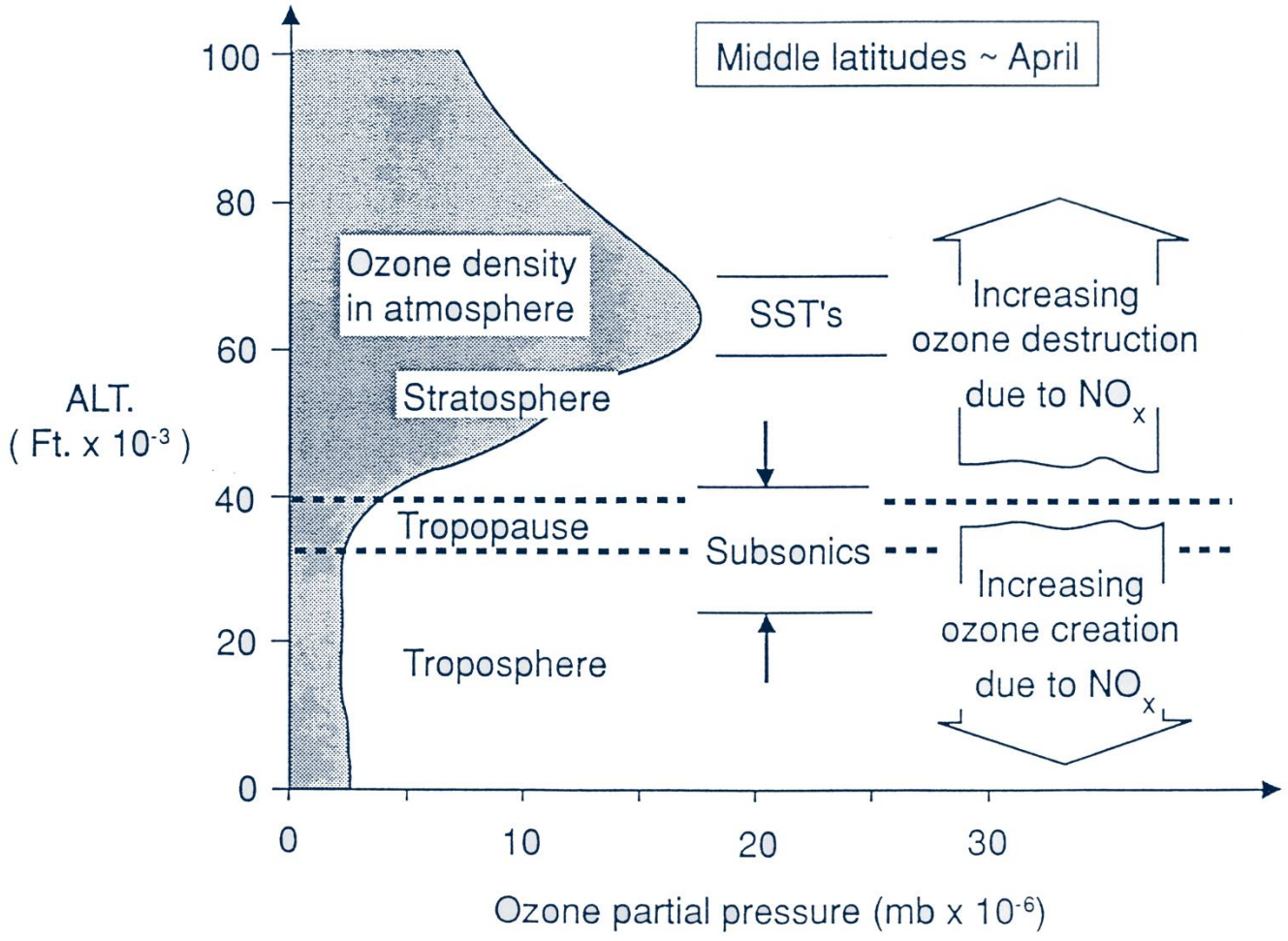
Aviation radiative forcing components (1940-2018)



LOSU = Level of scientific understanding

Source: Lee et al. (2020)

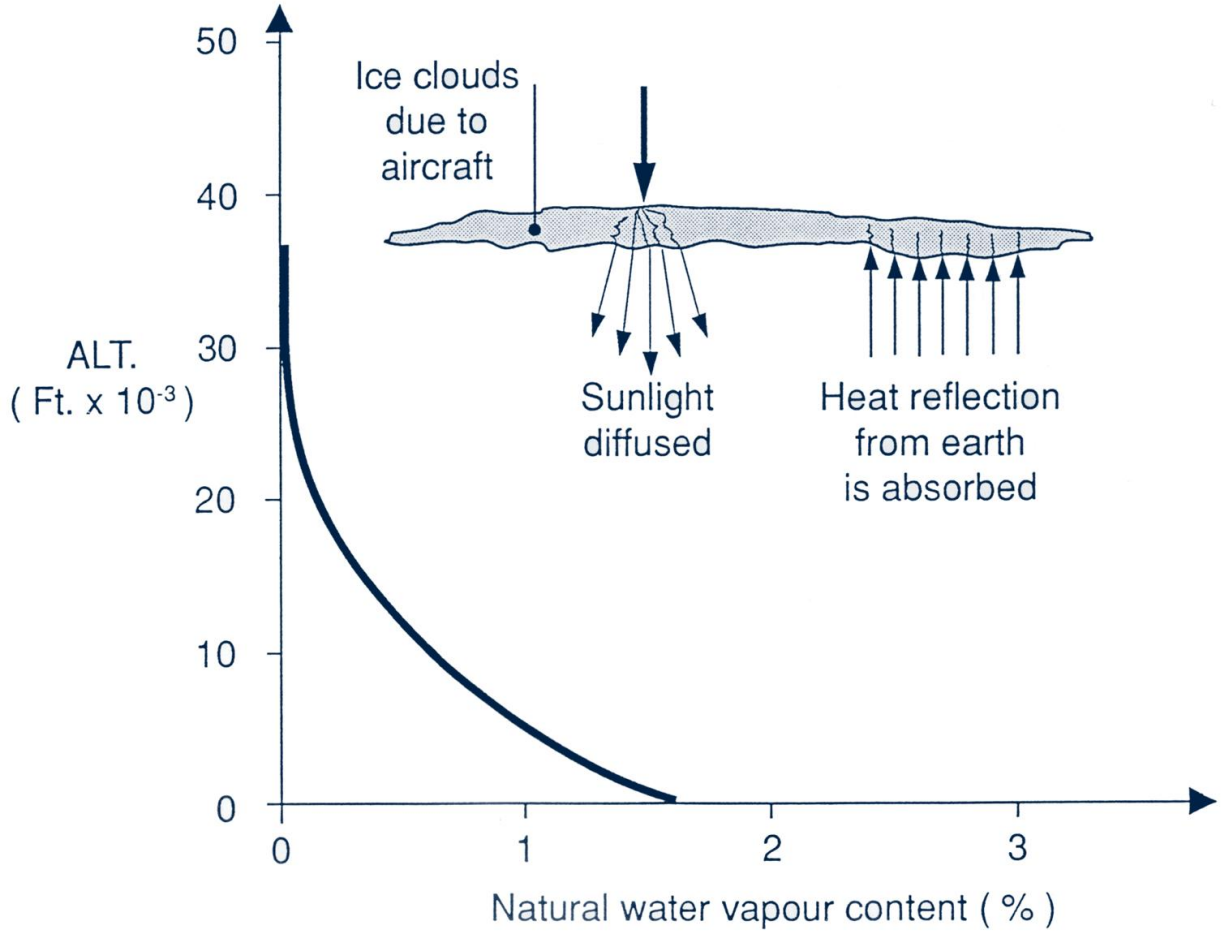
Effect of nitrogen oxides in the atmosphere



SST = Supersonic Transport

Source: Airbus

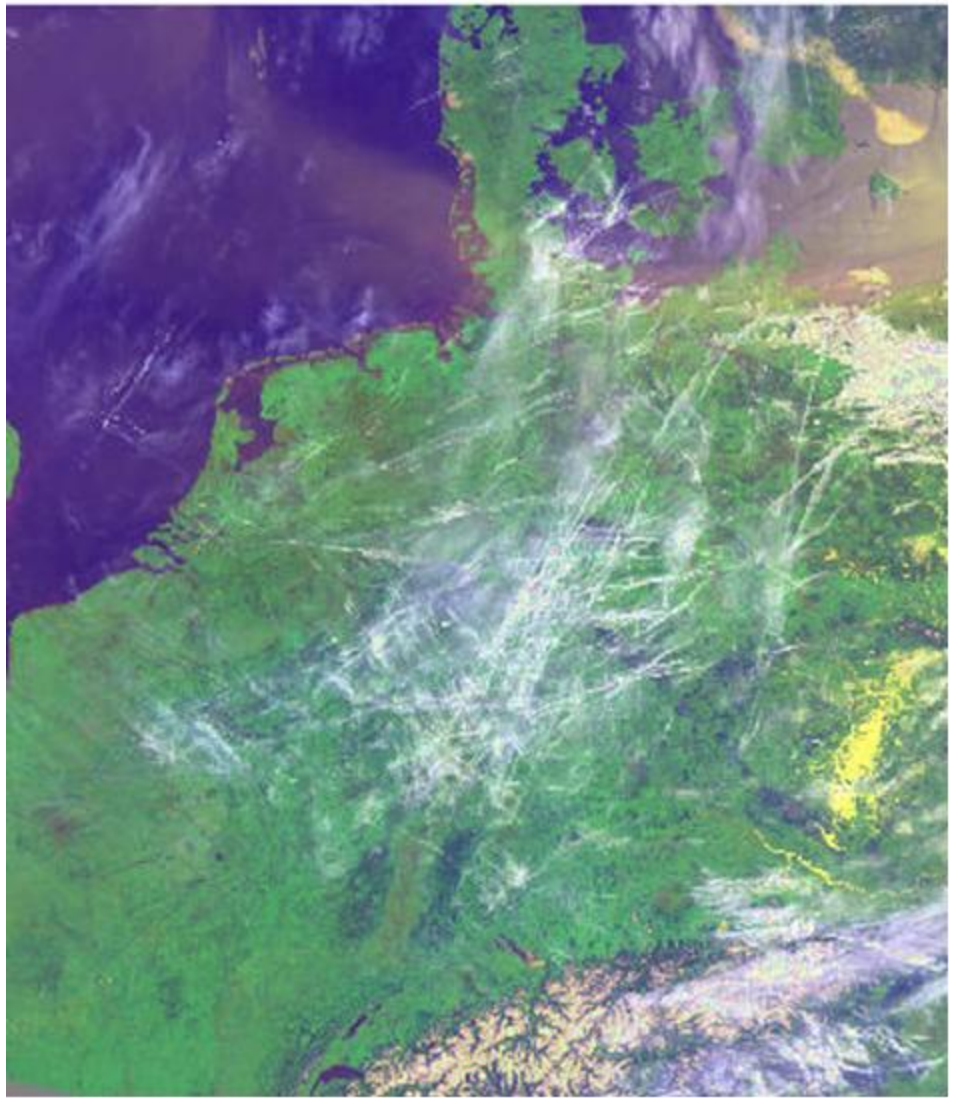
Effect of cirrus clouds in high altitudes



Depending on the **daytime**, the **season**, and the **altitude**, cirrus clouds have a cooling or warming effect.

Source: Airbus

Cirrus formation induced by contrails



The figure shows the formation of contrails in supersaturated air at an outside air temperature of below -40°C

- these are conditions that occur frequently in typical cruising altitudes of commercial transport aircraft (10 – 13 km)

Image source: DLR

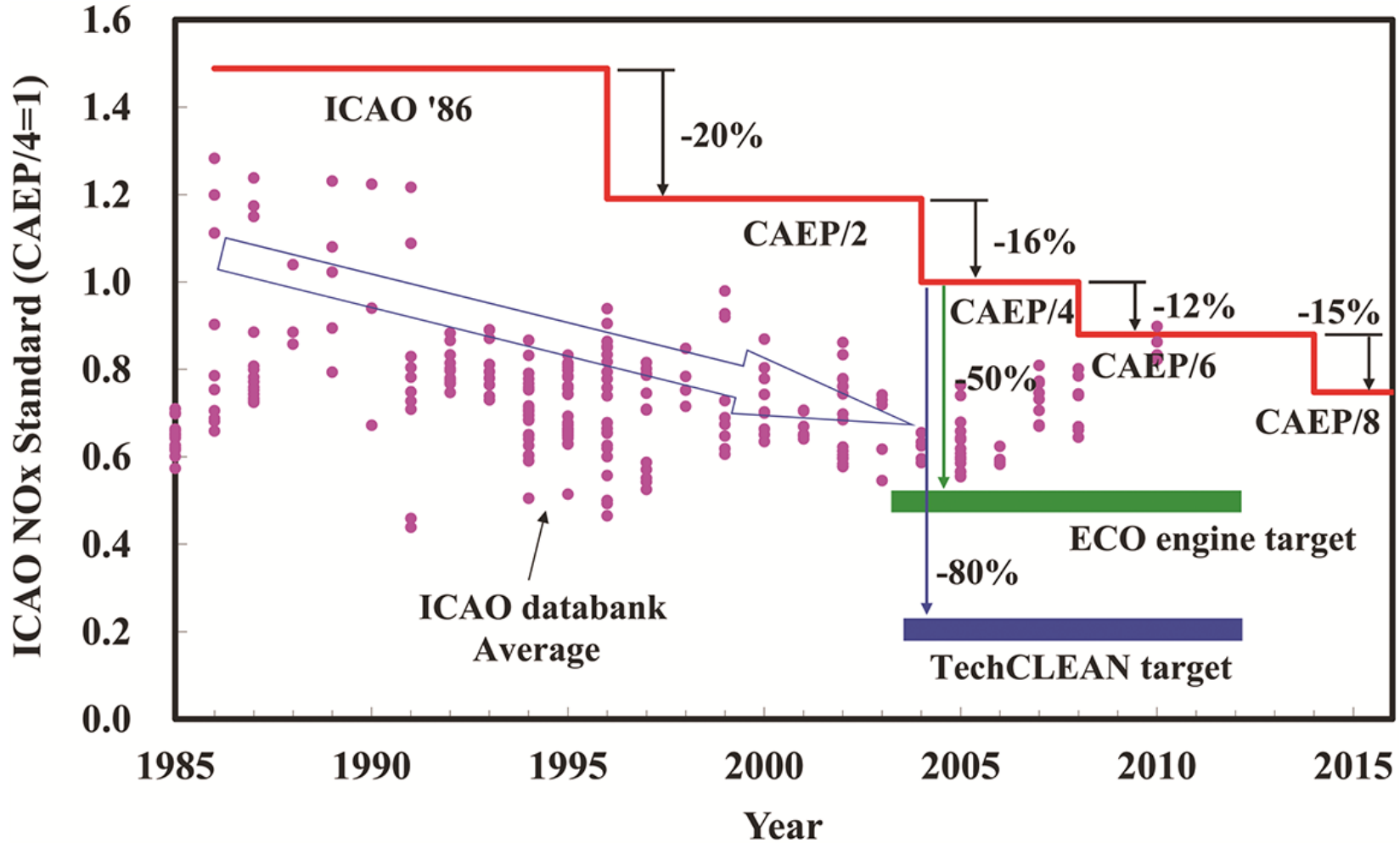
Means of environmental protection – Overview

- Limiting values for aircraft engine certification according to **ICAO Annex 16 Vol. II** for **NOx, CO, HC, and soot**. Since 1986, the limiting values have become more stringent in several steps.
- **Emissions-depending landings fees** in some European countries (i.e., Germany, UK, Denmark, Switzerland, Sweden as of 2012)
- Sweden introduced **emission fees** for CO_2 based on the LTO cycle in addition to the already existing landing fees
- **Environmental charge per passenger** introduced in the Netherlands
- Airlines suggest voluntary **CO_2-compensation charges** to passengers
- Integration of commercial aviation into the **European Emissions Trading System (EU-ETS)** since 2012
- Repeating political discussion about an introduction of a **kerosene tax**

ICAO Annex 16 Vol. II – Engine emission certification

- Applicable for all engines **with a rated output exceeding 26.7 kN** being produced after Jan. 01, 1983
- Excluded are turboprop, turboshaft, piston, and turbofan/turbojet engines (with a rated output below 26.7 kN)
- Emission measurements are conducted in dedicated test facilities. The measurement results are **referenced to MSL and ISA**
- The obeying of the limiting values is a **mandatory requirement of all ICAO member states**, which is examined by the national aviation authorities.
- The higher thermal degree of efficiency of modern engines leads to higher pressure ratios and temperatures in the combustion chamber. This again leads to an increased amount of NO_x emissions. In order to counteract this trend, the NO_x emissions-related limiting values have gradually been made more stringent.

Development of NO_x limiting values since 1986



Source: JAXA

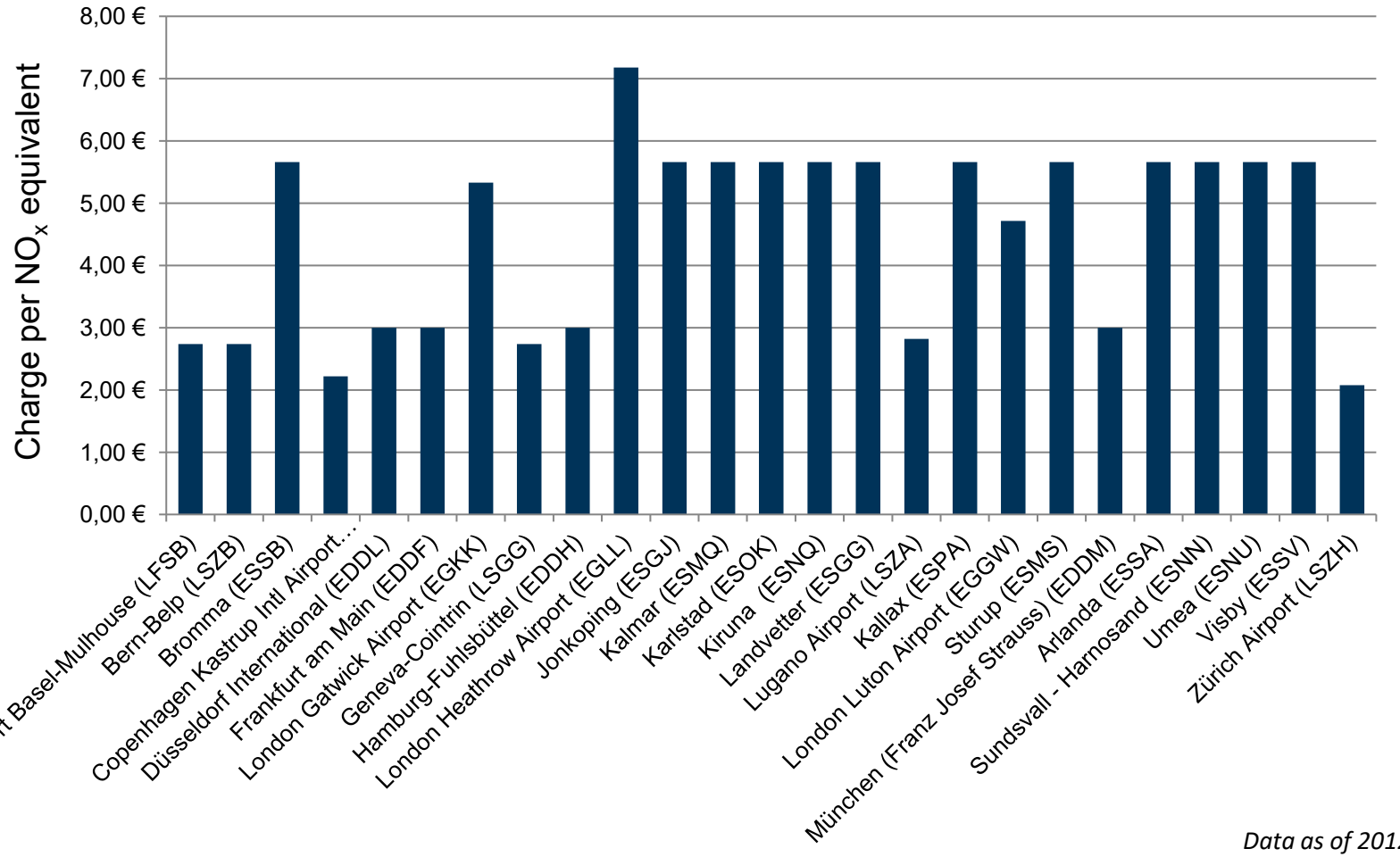
Emissions-depending landings fees (1)

	Airports	*System*
Switzerland (1997)	Geneva, Bern, Zurich, Lugano, Basel	Classification of aircraft into 5 emission categories depending on NO_X and HC emission output during the Standard ICAO LTO cycle.
Sweden (1998, 2004)	Bromma, Kalmar, Landvetter, Malmo, Stockholm	Common linear model based on ECAC 27-4 recommendation depending on the NO_X and HC emission output during Standard ICAO LTO cycle.
United Kingdom (2004)	London Heathrow, Gatwick	Individual authority decides whether or not aircraft below 8,618kg MTOW are included in the charge system.
Germany (2008)	Munich, Frankfurt, Cologne/Bonn, Hamburg, Dusseldorf	Each nation/airport has its own cost factor for one emission equivalent

Emissions-depending landings fees (2)

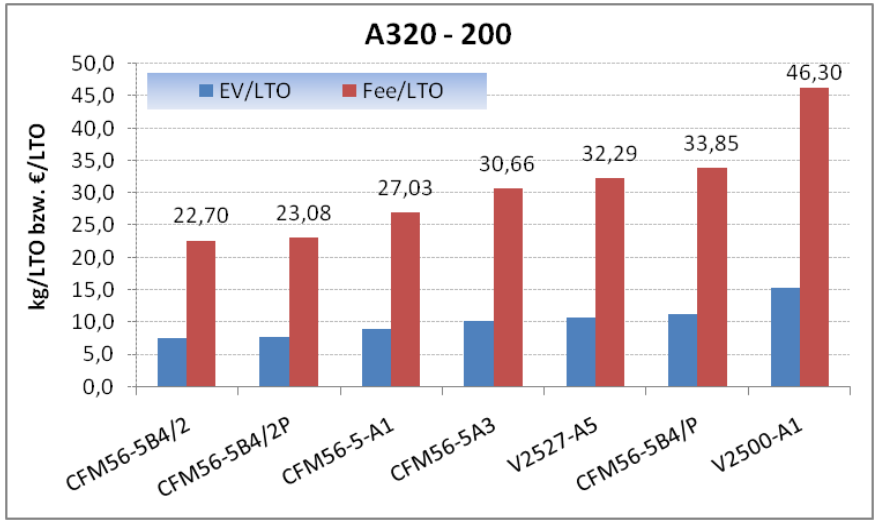
$$NO_x\ equivalent = no.engines \cdot a^* \cdot \sum_{LTO}(60 \cdot TIM \cdot FF \cdot EI(NO_x)^*\ /1000)$$

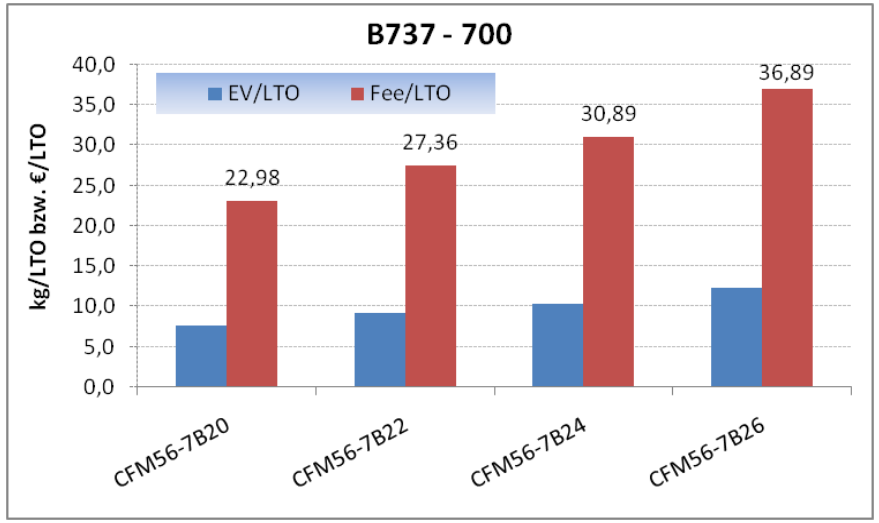
TIM = Time in Mode
FF = Fuel Flow
** = values from ICAO Emissions Databank*

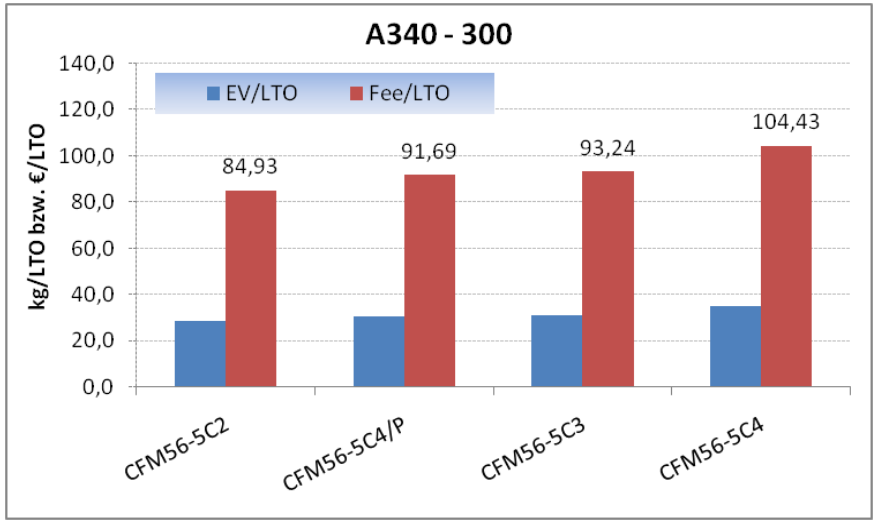


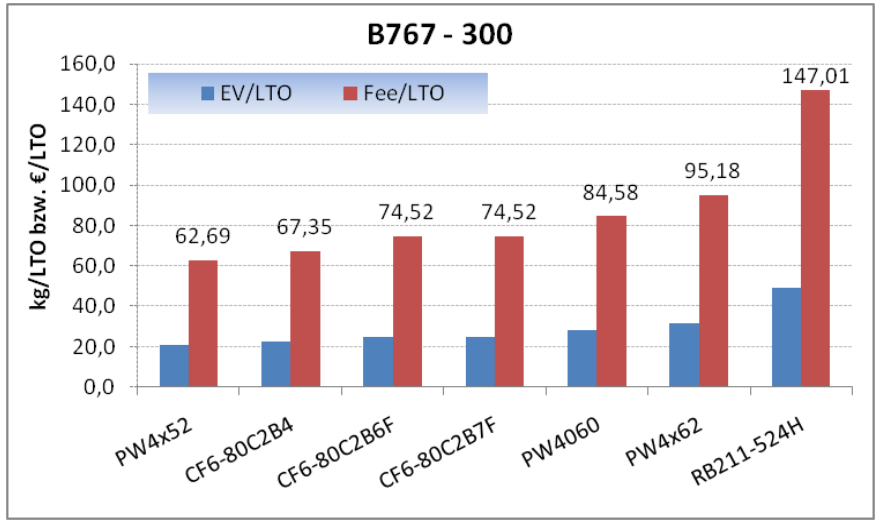
Data as of 2012

Emissions-depending landings fees: Examples (1)



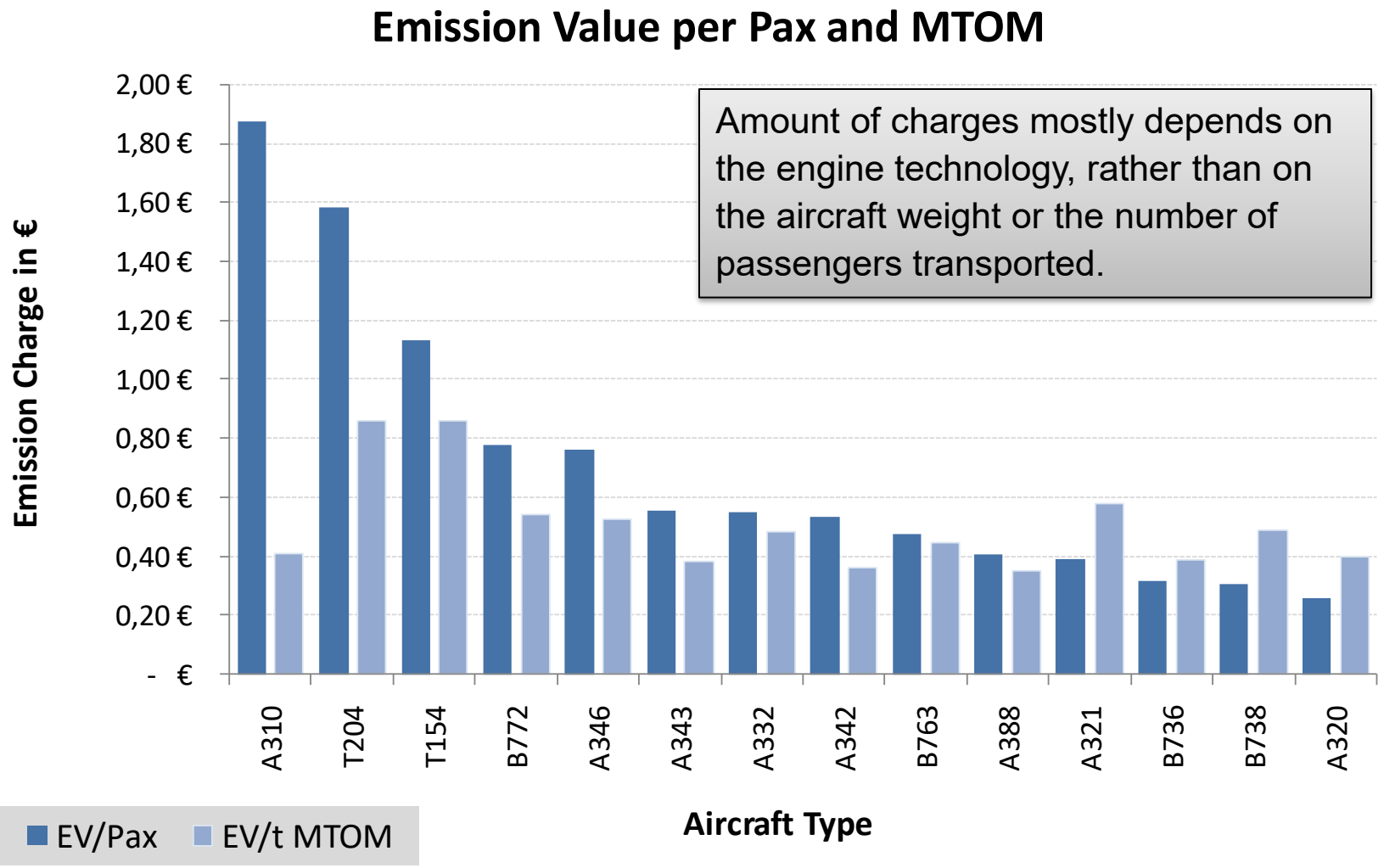






Data as of 2009

Emissions-depending landings fees: Examples (2)



Source: Munich Airport

European Emissions Trading System (EU-ETS)

The European Commission decided on October 24, 2008 **to integrate commercial aviation into the EU-ETS from 2012**. All flights were to be considered that arrive or depart from a EU airport.

- The system is **partially open**, i.e., aircraft operators are **allowed to buy CO_2-certificates from sectors other than aviation**
- **Only CO_2-emissions are taken into account**, no additional factors addressing other exhaust emissions (e.g., NO_x) are considered
- In 2012, the aviation sector receives certificates **representing 97%** of the average amount of CO_2-emissions produced yearly within the reference period of 2004 – 2006.
- This budget will be lowered to **95%** within the period of 2013 – 2020.
- The EU-ETS has caused **severe political discussions**: American and Asian governments threatened to ban their national carriers from complying with the scheme.

EU-ETS: Current State

- Prices for CO_2-certificates were **much lower than originally anticipated**
  - ➢ almost no incentive for airlines to reduce carbon emissions

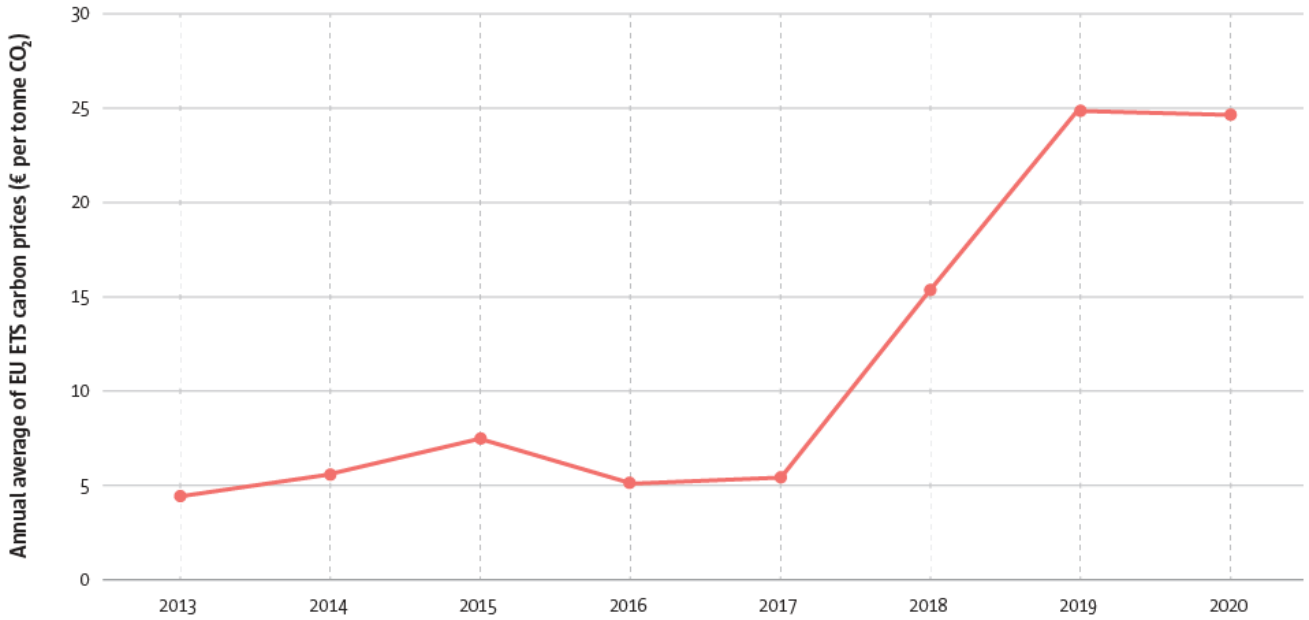
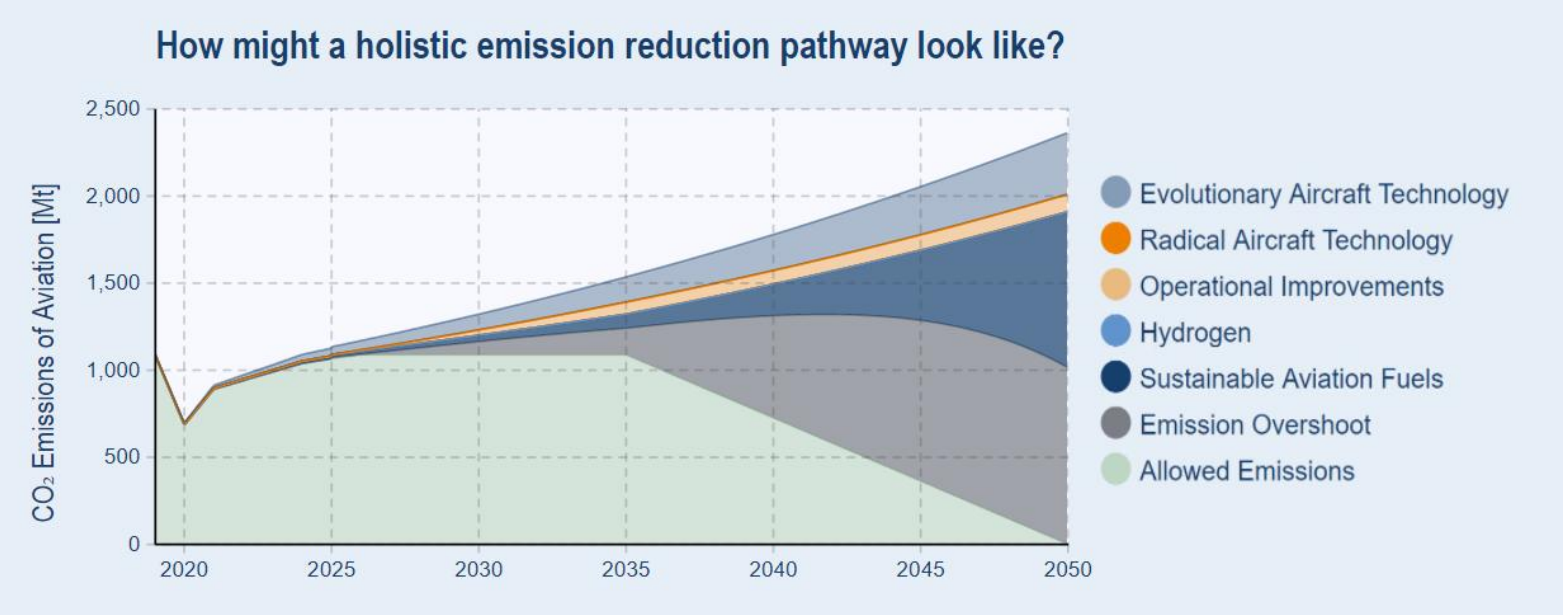


Image source:
easa - European Aviation Environmental Report

- Aircraft operator costs linked to purchasing EU Allowances have **gone up from around €84 milion (2013) to €955 milion (2019)**
  - Estimated to be **about 1.5% of total operating costs for aircraft operators** on flights within the scope of the EU ETS
- **Only European Domestic flights** (i.e., departure and arrival within the EU) are taken into account for the EU ETS

Reducing Aviation's Climate Impact



Aircraft Design

$$R = -\frac{L}{D} \cdot \frac{v}{g} \cdot \frac{1}{SFC} \cdot \ln\left(1 - \frac{m_{fuel}}{m_1}\right)$$

Aerodynamic Efficiency | **Propulsive Efficiency** | **Structural Efficiency**

Operations



Energy Sources



Reducing emissions: Engine

Decrease engine specific fuel consumption:

Ultra-high bypass ratio (e.g., CRISP)

Combustion chambers with low NO_x production:

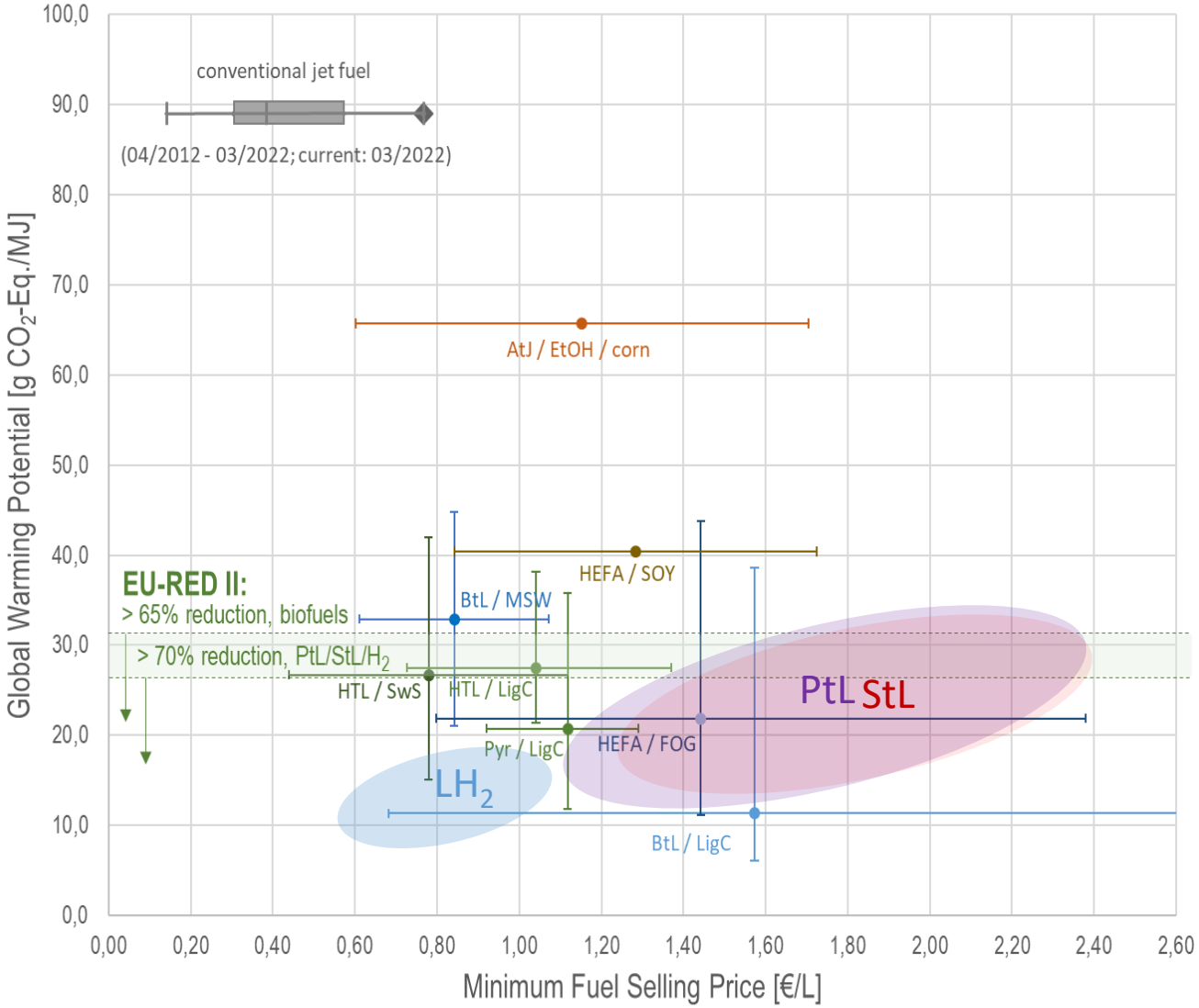
- LPP: Lean premixed prevaporized
- RQL: rich burn – quick quench – lean burn
- DAC: double-annular combustor

Optimization of the Joule-Brayton Process

Pratt & Whitney PW1000G



Reducing emissions: Sustainable Aviation Fuels

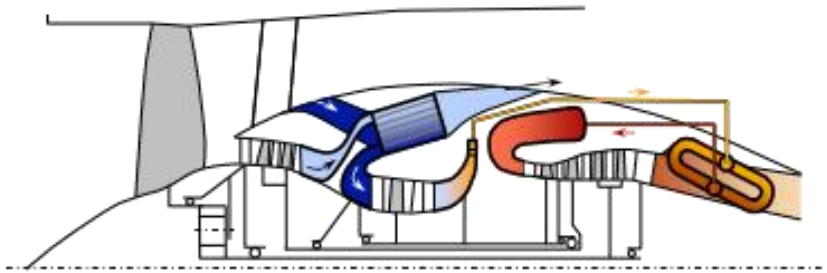


Source: Bauhaus Luftfahrt, 2023

Example: Intercooled Recuperated Core (1)

Description

- Intercooler located between low pressure and high pressure compressor
- Use of an exhaust heat exchanger (recuperator)
- Low overall pressure ratio @ ~25

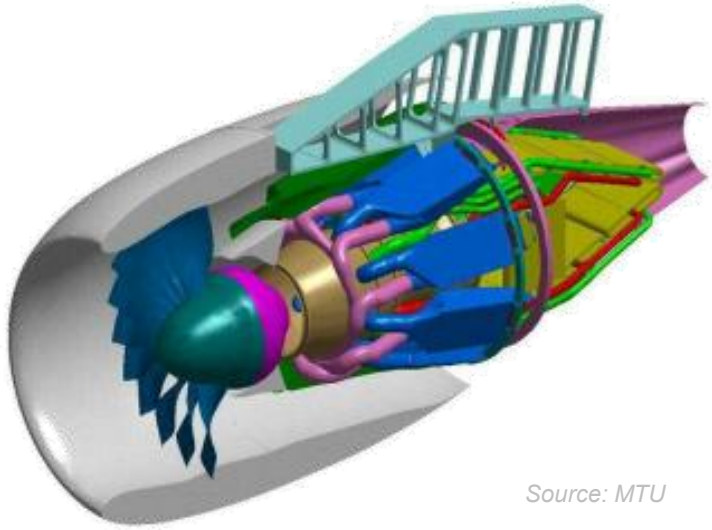


Advantages

- very high thermal degree of efficiency
- low NO_x emissions
- 30% fuel burn reduction estimated (rel. to Year-2000-engine)
- technological maturity expected from 2035

Technological Challenges

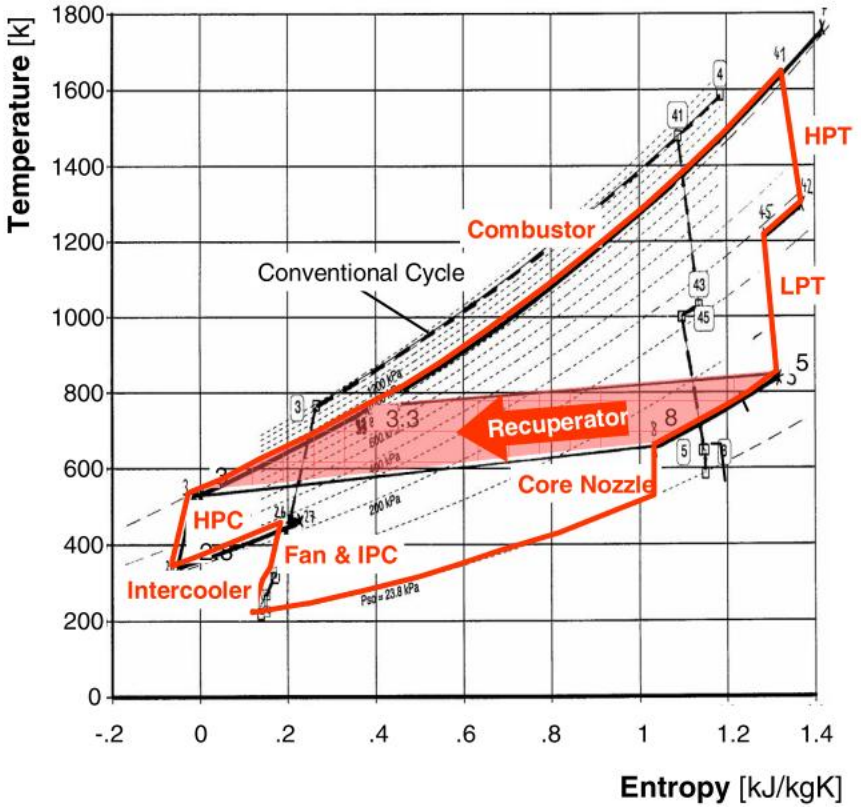
- Integration of intercooler and exhaust recuperator
- low pressure losses: air ducts, intercooler, recuperator
- light-weight and reliable system parts



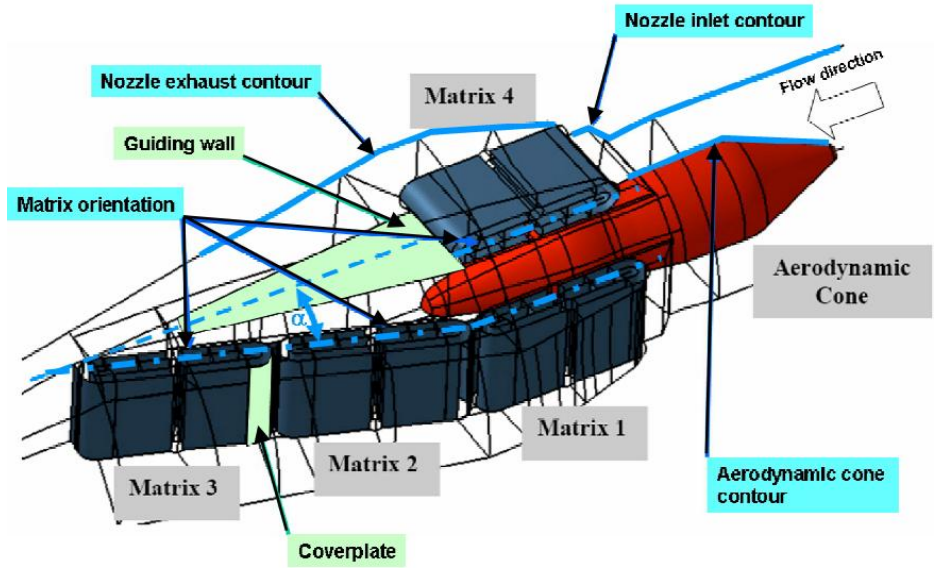
Source: MTU

Example: Intercooled Recuperated Core (2)

Thermodynamic cycle in the T-S Diagram



Position of the recuperator matrics



Source: MTU

Reducing emissions: Aircraft design

Aerodynamics: Reducing drag

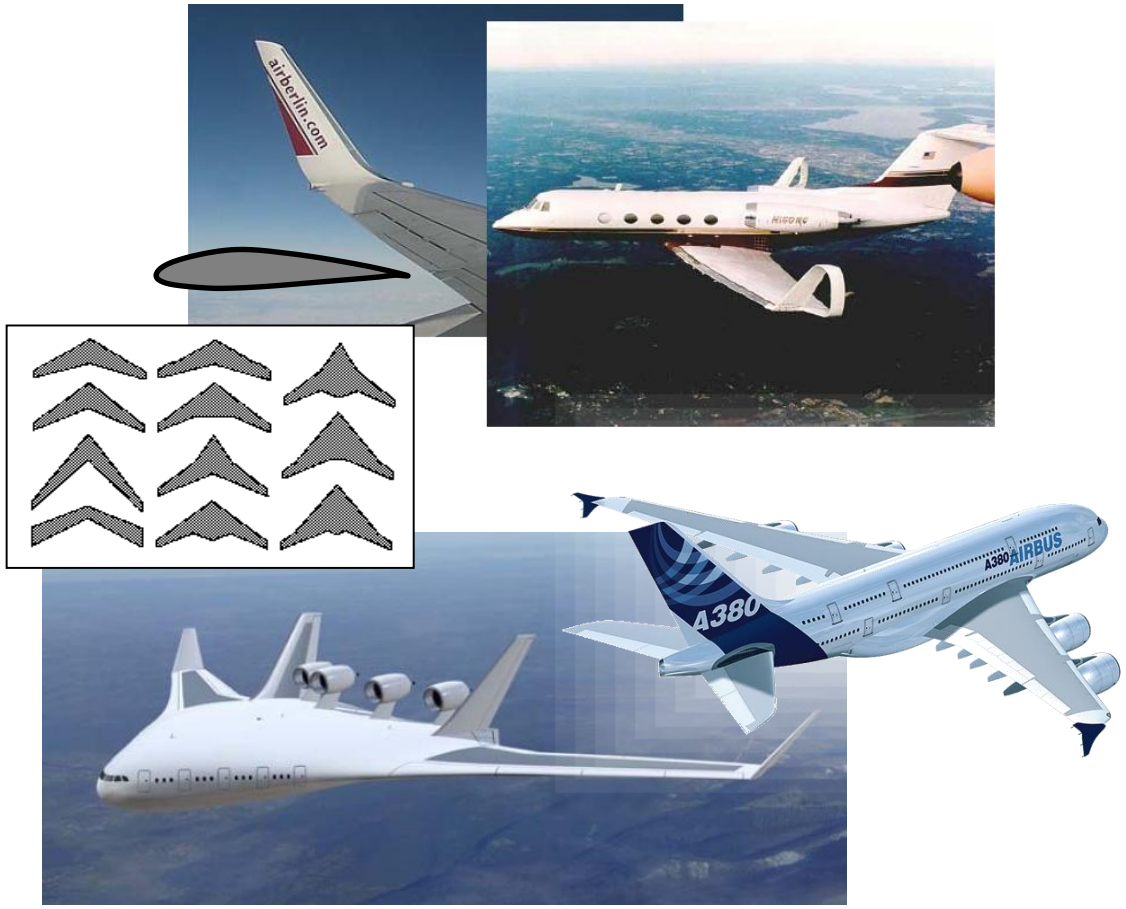
- Blended winglets, spiriods, riblets
- Advanced wing profiles
- Hybrid laminar flow control (HLFC)
- Adaptive wing

Structure: Reducing weight

- Materials: carbon composites, GLARE
- New construction methods

Configurations: Reducing weight and drag

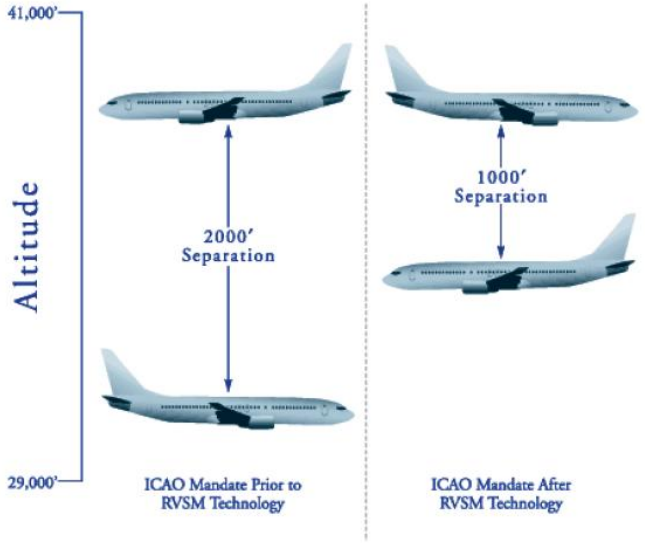
- Flying-wing configurations
- Three-surface aircraft
- Joined wing



Reducing emissions: Operations

Optimized Air Traffic Management

- Single European Sky
- Reduced Vertical Separation Minima (RVSM)
- Future Air Navigation System (FANS)
- Rerouting: avoid contrails

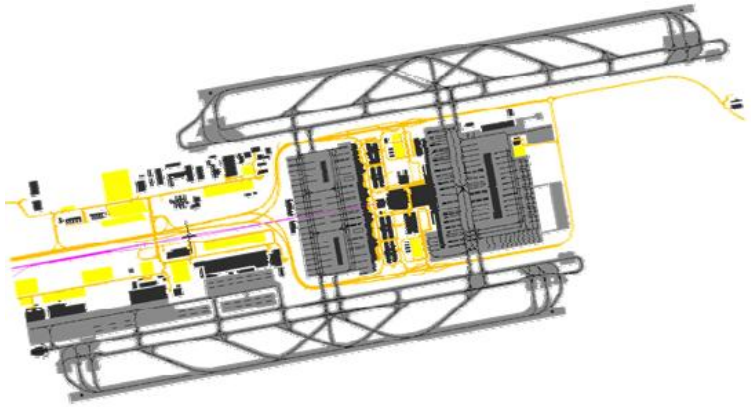


Low-emissions terminal procedures

- Continuous descent approach (CDA)
- Reduced-thrust takeoff

Low-emissions airport operations

- Use of low-emissions ground vehicles
- Advanced refueling systems
- Environmentally friendly airport access systems
- Optimized turn-around procedures
- Reduced / no-engine taxiing



Greener by Design …

Goal: Increase L/D

- Span ↑ Wing Weight ↑
  - light-weight materials & lower cruising Mach numbers: possible compensation

Radical Approaches:

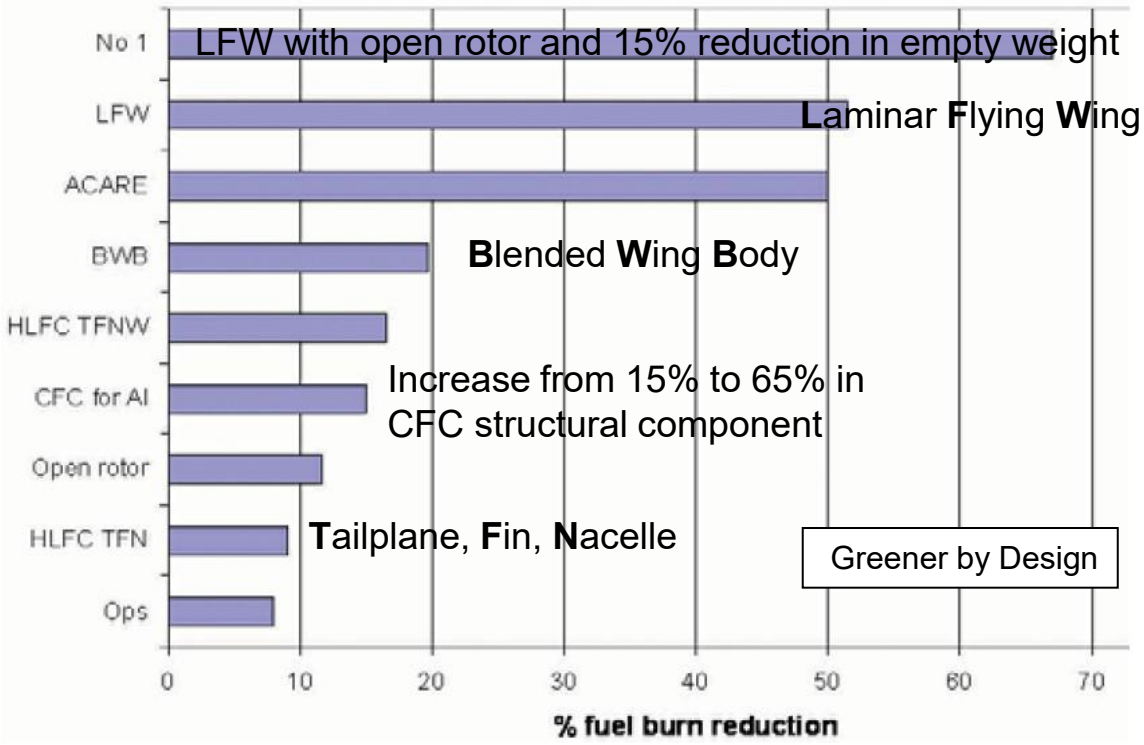
- Blended wing body
- Natural laminar flow control
- Hybrid laminar flow control
- Full (all-over) laminar flow control

Reference:

Aeronautical Journal, Aug 2006

Greener by Design, The Technology Challenge, 2001

Greener by Design, Mitigating the Environmental Impact of Aviation: Opportunities and Priorities, 2005



Source: Prof. Jeff Jupp: The impact of aviation on the environment – How will the future for air Transport be affected? RAeS lecture, Technische Universität München, 14 June 2007